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Numerical Parameter-Space Studies of Various Types of Thundercloud Gamma-Ray Emissions

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Key Points:

- Our numerical model, including relativistic feedback discharge, reproduces all known types of hard radiation signals from thunderclouds
- The model reproduces Gamma-Ray Glows and Terrestrial Gamma-ray Flashes given slowly and rapidly increasing electric fields, respectively
- Flickering Gamma-ray Flashes occur when the electric field increases at a moderate pace, given a large charge layer separation distance

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract

Until recently, mainly two types of hard radiation were observed in thunderstorms: intensive, short-lived terrestrial gamma-ray flashes (TGFs) and weak, long-lasting gamma-ray glows (GRGs). The Airborne Lightning Observatory for Fly's Eye GLM Simulator and TGFs (ALOFT, GLM: Geostationary Lightning Mapper) flight campaign revealed additional gamma-ray phenomena of intermediate characteristics, including flickering gamma-ray flashes (FGFs) consisting of multiple pulses with moderate intensity and duration. Here, we propose a model for reproducing hard radiation signals from thunderclouds, applying relativistic feedback discharge (RFD) theory and using a set of ordinary differential equations resembling a Lotka-Volterra predator-prey system. The model reproduces signals similar to the above-mentioned emission types, and a deeper parameter-space exploration reveals the following. Strong GRGs mainly result from an electric field building up slowly while remaining below the self-sustained feedback threshold, whereas TGFs are caused by a rapid jump in the electric field well beyond that threshold. FGFs serve as a middle ground resulting from the field increasing moderately until self-sustained feedback is barely attained. Compared with TGFs and GRGs, ALOFT's FGFs require a relatively large charge-layer separation distance to be produced, in agreement with their current status of being exclusively observed in large tropical thunderclouds. The simulations also indicate that the minimum charge layer distance needed to reproduce those FGFs can be reduced by allowing the electric field to increase slowly toward the minimum feedback threshold before rising at moderate pace. Our results indicate that RFD could play a significant role in several hard radiation phenomena produced in thunderstorms.

Plain Language Summary

Thunderclouds are known to emit gamma-ray emissions, both in the form of intensive, short-lived flashes and weak, long-lasting glows. The recent recordings of more intermediate types of radiation such as the flickering gamma-ray flashes, which appear as trains of pulses with moderate intensity and duration, indicate a more complex spectrum of different types of hard radiation in thunderstorms. Here, we present a new approach to reproducing that spectrum, including relativistic feedback discharge theory. Our results indicate that the model we developed is indeed capable of reproducing signals similar to almost any type of gamma-ray emission previously observed in thunderstorms. The above-mentioned flickering flashes are mainly reproduced in clouds with relatively large charge layer separation distances, compared to the other gamma-ray emission types, in agreement with the fact that they are currently observed exclusively in large tropical thunderclouds. We conclude that our results indicate that feedback discharges can play an important role in any type of gamma-ray emission originating from thunderclouds.

1. Introduction

Recent recordings of some very unique types of gamma-ray signals have challenged our current understanding of the physics within thunderstorms. Traditionally, hard radiation from thunderclouds has been divided into two clearly distinct classes: gamma-ray glows (GRGs, McCarthy & Parks, 1985; Parks et al., 1981), which typically last for a few seconds up to several minutes with a relatively low intensity, and terrestrial gamma-ray flashes (TGFs, Fishman et al., 1994), which last only for tens to hundreds of microseconds (sometimes up to a few milliseconds) but with intensities larger by several orders of magnitude. During the Airborne Lightning Observatory for Fly's Eye GLM Simulator and TGFs (ALOFT, GLM: Geostationary Lightning Mapper) flight campaign in July 2023, an intermediate type of gamma-ray emission, called flickering gamma-ray flashes (FGFs, Østgaard et al., 2024), was recorded. These FGFs are moderate in intensity and duration (compared to TGFs and GRGs) and consist of multiple gamma-ray pulses, lasting for 20–250 ms in total. The pulses are typically

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separated by 1–20 ms intervals, where the first few pulses tend to be longer (5–20 ms) and less intense, and the latter ones shorter (0.4–4 ms) and more intense. Whereas TGFs are generally accompanied by radio (Cummer et al., 2011; Dwyer & Cummer, 2013; Mezentsev et al., 2016; Stanley et al., 2006) and optical emissions (Østgaard et al., 2021; Skeie et al., 2022), indicating a possible connection to lightning leaders, FGFs appear radio and optically silent in observations, meaning that they are apparently not associated to lightning leader propagation. Currently, FGFs have only been observed in large tropical thunderclouds.

The same campaign also detected another phenomenon referred to as gamma-ray glow bursts (GGBs, Marisaldi et al., 2024), similar to GRGs but shorter and more intense, lasting only from 50 ms up to less than a second. They sometimes appear on top of GRGs and are often followed by FGFs. Furthermore, ALOFT recorded a whole population of weak TGFs that could not be observed with any space satellites, due to their low intensity (Bjørge-Engeland et al., 2024). These discoveries indicate the existence of an entire zoo of different gamma-ray phenomena produced in thunderclouds. Exploring that zoo is a crucial part of obtaining a more holistic understanding of the physical processes inside the high-electric-field region of a thunderstorm (referred to as the "high-field region" throughout the paper), as this could also ultimately affect our understanding of how lightning is initiated.

Today, it is well established that hard radiation in thunderstorms originates from runaway electrons, that is, electrons that have been accelerated by an electric field strong enough to ensure that the energy gain of the electrons exceeds the energy loss caused by scattering processes (Wilson, 1925). As these runaway electrons propagate through the thundercloud, they get decelerated by the electromagnetic fields of any gas atom nuclei that they closely pass by, leading to the emission of bremsstrahlung in the form of x-rays and gamma-rays. Furthermore, due to Møller scattering, any runaway electron may undergo avalanche multiplication on its way through the high-field region of the thundercloud, leading to a relativistic runaway electron avalanche (RREA, Gurevich et al., 1992; Luque, 2014). In this way, a single high-energetic seed electron injected into the high-field region can produce a large number of relativistic runaway electrons (RRE), up to a maximum of 10^5 in air, depending on the electric field and the total length of the avalanche propagation (Dwyer, 2007, see the discussion in section V concerning the maximum avalanche multiplication factor $\exp(\xi)$). The TGFs that are detectable from space have been estimated to require at least 10^{16} to 10^{17} RRE to be produced (Dwyer & Smith, 2005), which in turn requires at least 10^{11} seed electrons to be injected into the avalanche region when taking into account the maximum avalanche multiplication factor of 10^5. The source of such a high number of seed electrons is still under debate.

There are two leading candidates for mechanisms capable of accounting for the large number of seed electrons needed to produce TGFs. One is the relativistic feedback discharge (RFD) mechanism proposed by Dwyer (2003). It presumes that some of the photons emitted from the runaway electrons (as bremsstrahlung) undergo Compton backscattering or pair production. Thus, several of the produced photons and positrons will travel back to the start of the avalanche region and (through Compton and Bhabha scattering, respectively) release new seed electrons, which again may propagate through the avalanche region and trigger new RREAs. With this, a single seed electron (which may originate from a cosmic ray source) can trigger an avalanche of avalanches. Dwyer (2007) estimated the flux of energetic electrons (and consequently the flux of x-rays and gamma-rays) generated by this mechanism to exceed the corresponding flux generated by the standard RREA model (i.e., without feedback) by a factor of up to 10^{13}, limited only by the spatial charge which destroys the field. This makes the RFD model a promising candidate for explaining TGFs. The other candidate is the thermal runaway mechanism (Gurevich, 1960), which allows any electron to accelerate from thermal energy to runaway velocities given an electric field strength above a certain critical threshold. Such strong electric fields have not been measured on large scales in thunderclouds but may occur locally at the tips of streamers (Celestin & Pasko, 2011; Moss et al., 2006; Østgaard et al., 2016). The FGFs observed with ALOFT do not coincide with streamers or lightning leaders and are therefore unlikely to be driven by thermal runaway. Based on this, we use RFD theory as the basis for the model developed in this paper, since we here emphasize the ability to model as many as possible of the different types of gamma-ray emission detected in thunderclouds, including the FGFs observed with ALOFT.

Both RREA and RFD depend on the electric field strength of the high-field region lying above certain threshold values. The RREA threshold (Dwyer, 2003), that is, the electric field strength above which RREA can occur, is given by

$$E_{\mathrm{t}} = 284\ \mathrm{kV/m} \cdot \tilde{n}, \tag{1}$$

where $\tilde{n}$ denotes the ratio between the density of air at a given height and the corresponding density at sea-level. A representative reference height for thunderclouds is 12 km above sea level, where $\tilde{n} \approx 0.26$ (which is the value used in our models), meaning that one may expect RREA to typically occur in thunderclouds given an electric field strength above ~ 73 kV/m.

For relativistic feedback, there are two additional electric field thresholds, both lying slightly above the RREA threshold: (a) the minimum threshold E_f for feedback to occur; and (b) the self-sustained feedback threshold E_s. The latter threshold depends on the average vertical separation distance L_{cloud} between the positive and negative charge layers. Both thresholds can be found by first introducing the relativistic feedback factor, γ, defined as the ratio of the number of runaway electrons passing through the midplane of the high-field region after and before one feedback loop. Dwyer (2007, Equation 36) estimated this factor as follows,

$$\gamma \sim \exp\left(\left(\xi - \xi_0\right)\left(1 - \lambda/\lambda_a\right)\right), \tag{2}$$

where λ is the characteristic avalanche length, λ_a the attenuation length of the backward propagating seed producers (photons and positrons), $\xi \equiv L_{cloud}/\lambda$ the number of avalanche lengths in the high-field region, and ξ_0 the number of avalanche lengths when $\gamma = 1$. The avalanche length may be approximated by the following relation:

$$\lambda \approx \frac{U_R}{E - E_t}, \tag{3}$$

where $U_R = 7.3$ MV can be regarded as the feedback potential. Note that Dwyer (2007, Equation 13) used a value of 276 kV/m (multiplied by $\tilde{n}$) instead of the actual RREA threshold E_t in Equation 3 to fit the values for $1/\lambda$ (found through numerous Monte Carlo simulations) as a linear function of E. We use E_t instead, since λ actually goes to infinity as $E \to E_t$, and in this work we deal with electric fields close to E_t. For γ to be above unity, we need both $\lambda < \lambda_a$ and $\xi > \xi_0$. From the $\lambda < \lambda_a$ criterion, using Equation 3, one may deduce the minimum feedback threshold to be given by

$$E_f = E_t + \frac{U_R}{\lambda_a} = \left(284 \text{ kV/m} + \frac{U_R}{\lambda_{a,g}}\right) \cdot \tilde{n}, \tag{4}$$

where $\lambda_{a,g}$ is the ground-level (sea-level) attenuation length such that $\lambda_a = \lambda_{a,g}/\tilde{n}$.

In order to have self-sustained feedback, that is, more than 1 secondary seed is produced for each primary seed after a feedback loop ($\gamma > 1$), we must also require $\xi > \xi_0$. Defining the self-sustained relativistic feedback threshold E_s as the field strength ensuring $\xi = \xi_0$, one may rewrite the feedback factor as follows,

$$\gamma \sim \exp\left(\frac{(E - E_s)\,L_{cloud}}{U_R}\right). \tag{5}$$

The value of E_s, given any value of L_{cloud}, along with $\lambda_{a,g}$, on which E_f depends, can be found by performing a series of Monte Carlo simulations taking into account several cross sections. This was done by Dwyer (2007), and the black curve in Figure 3 of that article plots the value of E_s as a function of L_{cloud}, while E_f is the value to which the curve converges as $L_{cloud} \to \infty$. By curve-fitting that graph and rewriting Equation 2 as

$$\gamma = \delta \exp\left(L_{cloud}\left(\frac{1}{\lambda} - \frac{1}{\lambda_a}\right)\right), \tag{6}$$

the following estimates may be found:

$$E_f \approx 306 \text{ kV/m} \cdot \tilde{n}, \tag{7}$$

$$E_s \approx E_f + \log\left(\frac{1}{\delta}\right) \frac{U_R}{L_{cloud}} \cdot \tilde{n}, \tag{8}$$

where $\delta \equiv \exp(-\xi_0(1-\lambda/\lambda_a))$ is found to range from ~ 0.001 (for $L_{cloud} \lesssim 1$ km) to ~ 0.03 (for $L_{cloud} \gtrsim 2.5$ km). A more detailed derivation of the relation between δ and γ (for steady-state cases where $\gamma < 1$) may be found in Appendix C. Whenever the electric field in the high-field region surpasses E_s, so that $\gamma > 1$, the number of runaway electrons increases rapidly in time due to feedback. As a result of this, a large number of ions are produced as well. The positive ions will propagate backward through the high-field region, leading to partial or total quenching (discharging) of the electric field. The strong, rapid outburst of runaway electrons may, due to bremsstrahlung, result in a TGF. If the electric field amplitude gets only barely above, or barely below, this threshold before discharging, we might see a weaker signal in the form of an FGF or weak TGF, or possibly GRG/GGB, which is part of what we will explore in this paper. The above-mentioned electric field thresholds play a fundamental part of our analysis, since whether the electric field surpasses each of these thresholds, and how fast it approaches the thresholds, can be crucial for which types of hard radiation signals that are emitted from the thundercloud.

One major goal of this paper has been to develop a new model, which applies RFD theory, capable of reproducing FGFs akin to those detected by ALOFT, along with other well-known gamma-ray emissions from thunderclouds, such as GRGs and GGBs. We even examine the capability of this RFD-based model to reproduce (single- or multipulse) TGFs, despite the observational fact that those features are commonly associated with either leaders or streamer bursts, which we do not explicitly model here. However, as our model applies an external ad hoc electric field when reproducing any hard radiation signal through RFD, the external electric field source can be interpreted as a contribution from leaders/streamers in the cases where a TGF is reproduced. It is also possible that RFD could be responsible for producing a vast amount of weak TGFs not seen with space satellites (such as those recently discovered with ALOFT) even in the absence of leaders or streamers. Because of this, we here aim to examine to which extent this RFD-based model can work as a universal model for reproducing hard radiation from thunderclouds.

Furthermore, we aim to explore the parameter space of this model by comparing the results of simulations with various input values for the following parameters: the separation distance between the positive and negative charge layers, the amplitude and the characteristic time scale of an external electric field source, the initial electric field strength, and the ion attachment length. With this parametric study, we gain insight into how large ranges of values our model can apply for each of those parameters in order to reproduce any of the aforementioned gamma-ray signals observed in thunderstorms. We will also study how the electric field evolves with respect to the thresholds E_t, E_f, and E_s as the cloud is charged and eventually discharged according to the RFD mechanism and especially localize for which cases self-sustained feedback is achieved. The paper is structured as follows. In Section 2, we describe our model and methodology. In Section 3, we go through the results of our simulations and parameter-space analysis. Finally, in Section 4, we conclude by discussing the implications of our findings for the RFD theory, discussing the limitations of our model, and proposing directions for future research.

2. Methods

We propose a model based on a system of ordinary differential equations (ODEs) which describes the internal electric field (built up by cloud charges) E_{cloud} and the number density of RRE and ions (n_R and n_{ion}, respectively) in the high-field region of a thundercloud. The system is approximated to be uniform in space, but its vertical size L_{cloud} is explicitly included in order to describe the RREA multiplication. Therefore, this model can be regarded as "0.5-dimensional," or 0.5D for short.

In our model, the high-field region is approximated to consist of two equally oppositely charged parallel plates separated by the vertical length L_{cloud}. A sketch of this region and its total electric field $\mathbf{E}$ is shown in Figure 1. Seed electrons are assumed to be drawn toward the positive-polarity plate and accelerated to relativistic velocities by the electric field, hence becoming runaway electrons. In computing the time derivatives of the ion and runaway electron densities, the model implements the RREA mechanism by assuming that the runaway electrons undergo Møller scattering with atomic electrons several times before reaching the positive-polarity plate, thus producing secondary electrons. Furthermore, RFD is implemented by assuming that some of the bremsstrahlung (produced by runaway electrons colliding with nuclei) undergo Compton backscattering or pair production, which eventually results in some photons and positrons traveling back toward the negative-polarity plate, where they produce new seed electrons (through Compton and Bhabha scattering) that create new electron avalanches.

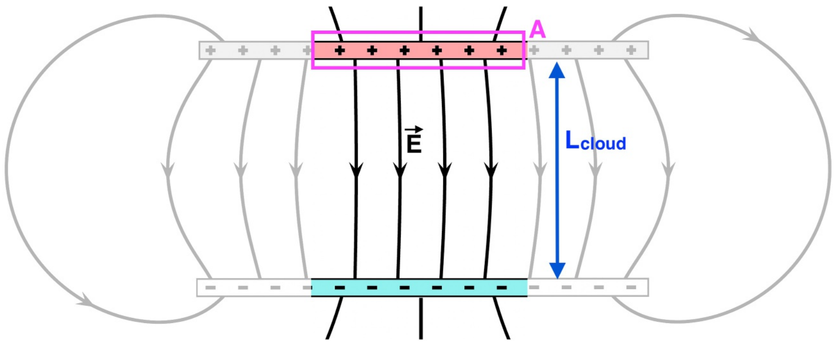

Figure 1. Sketch of the high-field region of a thundercloud, approximated by two parallel plates of opposite polarity separated by the distance L_{cloud}. The colored part marks the central part of the high-field region, where the electric field $\mathbf{E}$ may be approximated to be vertical and homogeneous. This field consists of an internal component $\mathbf{E}_{\mathrm{cloud}}$, calculated self-consistently from the density of cloud charges, and an externally imposed field $\mathbf{E}_{\mathrm{ext}}$, given as an ad hoc term. $\mathbf{A}$ marks a thin prism-shaped surface area that encloses the colored part of the positive-polarity plate, used to calculate the charge moment later in this paper.

In order to simulate the thundercloud building up its electric field until exceeding the feedback thresholds, we assumed that the high-field region is exerted to an external field source E_{ext}, in our case given by a sigmoid function with half-width τ rising from zero toward a constant value $k_{\mathrm{E}}E_{\mathrm{f}}$, where k_{E} characterizes the relative amplitude of the external electric field contribution. With this, the total electric field of the cloud is given by

$$E = E_{\mathrm{cloud}} + E_{\mathrm{ext}}. \tag{9}$$

It is important to note that our model does not say anything about where the external field contribution E_{ext} comes from. It can be due to a lightning leader, or just due to the convective currents inside the cloud, or something else. Initially, the electric field is assumed to be given by a background field E_0, which in our cases is set slightly above the RREA threshold E_{t} (by default $E_0 = 1.04E_{\mathrm{t}}$). In the real case, the thundercloud would start with an electric field strength well below E_{t} and slowly charge up until it surpasses that threshold. Since we expect any detectable hard radiation to occur first after E surpasses E_{t}, we chose to start the simulation at a point where $E > E_{\mathrm{t}}$ in all our cases.

As E_{ext} rises from zero toward $k_{\mathrm{E}}E_{\mathrm{f}}$, the total electric field rises from E_0 toward the self-sustained feedback threshold E_{s}, until discharging due to the self-quenching effects caused by the cloud conductivity enhanced by relativistic feedback or, in some cases, discharging even before these self-quenching effects kick in, eventually leading to the production of TGFs, FGFs, GRGs, or GGBs (or a combination of those). Since this discharge occurs shortly after E surpasses E_{s}, meaning that negative contributions from E_{cloud} cancel out positive contributions from E_{ext} (until $E < E_{\mathrm{s}}$ again), the total external contribution $k_{\mathrm{E}}E_{\mathrm{f}}$ can be much larger than E_{s} without causing the total electric field to go far beyond that threshold. Thus, we allowed k_{E} to be anything from 0.03 to 6 (see Section 2.3) in our simulations.

As we want our simulation results to stay within a realistic frame, one criterion used when choosing input values for the free parameters of our model (as listed in Section 2.3) is to ensure that the charge-moment changes do not exceed far beyond the largest values measured in any lightning events on Earth. The charge moment change Δp to which the high-field region is exerted as a consequence of the external field source can be written as

$$\Delta p = \Delta Q L_{\mathrm{cloud}}, \tag{10}$$

where ΔQ denotes the charge moved during the increase in the electric field in the high-field region. A minimum value for the charge moment change can be found by estimating the charge moved along the central part of the high-field region, as sketched in Figure 1. Let $\mathbf{A}$, as marked in the figure, be the prism-shaped surface area that encloses the part of this region that surrounds the positive-polarity plate. The total charge Q enclosed by $\mathbf{A}$ can then be estimated through Gauss' law as

$$Q = \epsilon_0 \oint \mathbf{E} \cdot \mathrm{d}\mathbf{A}, \tag{11}$$

where ϵ_0 is the vacuum permittivity. Since the horizontal size of the cloud is assumed to be significantly larger than the vertical size, one may set the horizontal side lengths of **A** equal to L_{cloud} to obtain the following minimum estimate for the charge enclosed by **A**:

$$Q_{\min} \sim \epsilon_0 E L_{\text{cloud}}^2. \tag{12}$$

Putting this into Equation 10, we find that the minimum charge moment change due to the external electric field contribution $\Delta E_{\text{ext}} = k_{\text{E}} E_{\text{f}}$ can roughly be estimated as

$$\Delta p_{\min} \sim \epsilon_0 k_{\text{E}} E_{\text{f}} L_{\text{cloud}}^3. \tag{13}$$

In our model, we allow L_{cloud} to have an upper limit of 12 km (see Section 2.3), which is possibly in the very upper range of charge layer separation distances attainable in any terrestrial thunderclouds. Combining this with our chosen upper limit of k_{E} (set to 6), we find that the above-given estimate for the minimum charge moment change due to the external field may range up to 7300 C km, which is roughly one order of magnitude above the charge moment change measured in the biggest sprite-producing lightnings seen on Earth (e.g., see Li et al., 2012, which measures charge moment changes up to 600–1000 C km). However, one may also find that the charge moment change will remain roughly below 1000 C km as long as $k_{\text{E}} \leq 6$ for $L_{\text{cloud}} \leq 6$ km or $k_{\text{E}} \lesssim 1$ for $L_{\text{cloud}} = 12$ km. Hence, with the above-given upper limits for L_{cloud} and k_{E}, we avoid unrealistically large charge moment changes for most of our cases.

In the following section, we will go through the ODEs that form the basis of our 0.5D model. Furthermore, we will delve into our methodology for determining which type of signals that are reproduced from the model given a certain set of parameters. Finally, we will have a look at which parameters we decided to explore systematically in our parameter-space analysis.

2.1. Model Equations

Our model equations for E_{cloud}, n_{R}, and n_{ion} are derived from Gauss' law with the assumption of $E_{\text{cloud}} = 0$ outside upper and lower boundary of the high-field region along with continuity equations for RRE and ions given spatial uniformity (hence no divergence terms) and source terms derived from RFD theory (as outlined by Dwyer, 2007), hence taking the shape of ODEs on the following form,

$$\frac{\mathrm{d}E_{\text{cloud}}}{\mathrm{d}t} = \frac{J_0 - (J_{\text{R}} + J_{\text{sec}} + J_{\text{ion}})}{\epsilon_0}, \tag{14}$$

$$\frac{\mathrm{d}n_{\text{R}}}{\mathrm{d}t} = \frac{(\gamma' - 1)\, n_{\text{R}} + n_{\text{R0}}(E)}{\tau_{\text{round}}}, \tag{15}$$

$$\frac{\mathrm{d}n_{\text{ion}}}{\mathrm{d}t} = -\eta_{\text{ion}} n_{\text{ion}} + S_{\text{ion}} + S_{\text{ion}}^{\text{cloud}}, \tag{16}$$

where the dependency of the vertical extent of our domain implicitly lies in the right-hand-side terms of Equation 15, hence making this a 0.5D problem. In addition to ions and RRE, the production of low-energy electrons is also taken into account by including a secondary-electron current density term (J_{sec}) in Equation 14. The right-hand-side terms of Equations 14–16, along with the quantities they depend on, are defined below.

- $J_0 = \sigma^{\text{cloud}} E_0$ is the cloud charging current needed to maintain the background (initial) electric field E_0, given the cloud's background conductivity σ^{cloud}, which we parametrize using a Maxwellian relaxation time $\tau^{\text{cloud}} \approx 1$ s so that

$$\sigma^{\text{cloud}} = \frac{\epsilon_0}{\tau^{\text{cloud}}}. \tag{17}$$

- $J_R = en_R v_R$ is the RRE current, given the elementary charge e and the RRE drift velocity (Coleman & Dwyer, 2006),

$$v_R = 0.89c\ . \tag{18}$$

- $J_{sec} = en_{sec} v_{D,e}(E)$ and $J_{ion} = en_{ion} v_{D,ion}(E)$ denote the secondary electron and ion currents, respectively, given the secondary electron number density n_{sec} (see below) and the electron/ion drift velocities,

$$v_{D,e}(E) = \mu_e(E)E \quad \text{and} \quad v_{D,ion}(E) = \mu_{ion}E, \tag{19}$$

with the electron/ion mobilities (Elford & Rees, 1974; Lehtinen, 2020) given by

$$\mu_e(E) = \left(\frac{E_g}{3\ \mathrm{MV/m}}\right)^{-0.17} \times 0.044\ \mathrm{m^2V^{-1}s^{-1}} \cdot \tilde{n}^{-1} \tag{20}$$

$$\mu_{ion} = 2.3 \times 10^{-4}\ \mathrm{m^2V^{-1}s^{-1}} \cdot \tilde{n}^{-1} \tag{21}$$

where $E_g \equiv E/\tilde{n}$. Since n_{ion} denotes the total density of ions, J_{ion} includes contributions both from the positive and the negative ions. Since they are oppositely charged as well as moving in opposite directions, their current contributions end up pointing in the same direction.

- The secondary electron number density is given by $n_{sec} = S_{ion}/\nu_{att}(E)$.
- $S_{ion} = \alpha_{ion} v_R n_R$ is the production rate (in $\mathrm{m^{-3}s^{-1}}$) of ion-electron pairs from RREA.
- α_{ion} is the ion-electron pair production (per meter) by RRE. This value may be found in Dwyer and Babich (2011) to be equal to 8350 $\mathrm{m^{-1}}$ at zero altitude. Interestingly, we also can get a close estimate for it from the widely used ion-pair production energy by energetic particles in air $\Delta\varepsilon_{ion} = 35$ eV (Rees, 1989, p. 40) and the field at which the energy losses equal the gains, which may be taken to be E_t, so that $\alpha_{ion} \approx E_t/\Delta\varepsilon_{ion}$, which is the formula we used for this term in our model (taking into account its scaling with air density).
- $\nu_{att}(E)$ is the electron attachment rate (in $\mathrm{s^{-1}}$), given by the electron drift velocity multiplied by a sum of a two-body term and a three-body term (Lehtinen, 2020, and references therein). The two-body term is given by Townsend approximation,

$$\alpha_{a2}(E) = \alpha_{a0} \exp\left(-\frac{E_{a0}}{E}\right), \tag{22}$$

with $\alpha_{a0} = 1.7 \times 10^3\ \mathrm{m^{-1}}$ and $E_{a0} = 3.2$ MV/m. The three-body term is given by

$$\alpha_{a3}(E) = A_{a3} \cdot n_L^2 \left(\frac{E}{n_L} \cdot B_{a3}\right)^{-1.2749}, \tag{23}$$

where the Loschmidt constant $n_L = 2.688 \times 10^{25}\ \mathrm{m^{-3}}$ denotes the average density of neutral air at sea level, and the remaining constants are given by $A_{a3} = 4.7778 \times 10^{-69}\ \mathrm{m^5}$ and $B_{a3} = 10^4\ \mathrm{V^{-1}m^{-2}}$. The two-body term scales linearly with the air density, and the three-body term quadratically with the air density, so that the total electron attachment rate can be written as

$$\nu_{att}(E) = v_{D,e}(E)\left[\alpha_{a2}(E_g) \cdot \tilde{n} + \alpha_{a3}(E_g) \cdot \tilde{n}^2\right]. \tag{24}$$

- γ' denotes a corrected feedback factor, adjusted in order to give correct values of n_R both for steady-state ($\gamma < 1$) and growing ($\gamma \geq 1$) cases. It is given by $\gamma' = 1 + \log\gamma$ for $\gamma \geq 1$ and $\gamma' = \gamma$ for $\gamma < 1$, with γ given by Equation 5 where the self-sustained feedback threshold E_s is given by Equation 8 and set to match the values found by Dwyer (2007) for any L_{cloud}. This way, our feedback factor incorporates results from 3D Monte Carlo simulations (assuming trasverse size much larger than longitudinal size, however), which even takes into account feedback contributions from RRE outside the high-field region.

- $n_{\text{R0}}(E)$ is the value n_{R} would have if no feedback, for example, given a cosmic ray source term S (in our cases, we assumed a constant cosmic ray flux, given by $S = 10\ \text{m}^{-3}\text{s}^{-1}$), we have

$$n_{\text{R0}}(E) = S\frac{L_{\text{cloud}}}{v_{\text{R}}}\frac{\exp(y(E)) - 1}{y(E)}, \quad y(E) \equiv (E - E_{\text{t}})\frac{L_{\text{cloud}}}{U_{\text{R}}}. \tag{25}$$

- $\tau_{\text{round}} = 2L_{\text{cloud}}/v_{\text{R}}$ is the round-trip time for one feedback loop.
- $\eta_{\text{ion}} = v_{\text{th,ion}}(E)/\Lambda_{\text{ion}}$ is the absorption rate of ions (in s^{-1}) due to their attachment to hydrometeors.
- $v_{\text{th,ion}}(E)$ denotes an approximate effective ion velocity including a thermal component,

$$v_{\text{th,ion}}(E) = \sqrt{\frac{k_{\text{B}}T_{\text{cloud}}}{M_{\text{ion}}m_{\text{u}}}} + \mu_{\text{ion}}E, \tag{26}$$

given the Boltzmann constant k_{B}, cloud temperature $T_{\text{cloud}} = 200$ K, mean molecular weight (of NO_3^- ions) $M_{\text{ion}} = 62$, and atomic mass unit $m_{\text{u}} = 1.66 \times 10^{-27}$ kg.

- Λ_{ion} is the ion attachment length, that is, the mean free path that ions travel before attaching themselves to water droplets, which lies typically in the range of $3 - 20$ m (Dwyer, 2005).
- $S_{\text{ion}}^{\text{cloud}}$ is the background ion source (in $\text{m}^{-3}\text{s}^{-1}$) of the cloud, which sustains the background cloud conductivity (see Appendix A).

All simulations in this paper were run by numerically solving Equations 14–16 over a time interval from t_1 to t_2, where t_1 typically characterizes a time well before the change in E occurs, and t_2 a time well after that change. For each case, depending on the chosen increase time scale τ of the external electric field, we used $t_1 = -10\tau$ and $t_2 = 10\tau$. One may note that our model equations bear resemblance to a system of Lotka-Volterra predator-prey equations (see Appendix B), which indeed has solutions in the form of spontaneous current oscillations due to the coupling between the current and the electric field. Because of this, features that resemble FGFs are highly expected to be simulated from our model equations.

2.2. Methodology for Studying and Classifying the Signals Reproduced in the Simulations

Classifying gamma-ray signals is done mostly by measuring their intensities, durations, and frequencies. Synthesizing intensity curves for gamma-ray emissions from runaway electrons will be implemented as part of one of our future papers. We expect this intensity to be proportional to n_{R}. Therefore, we may simply study the plots of n_{R} against time, as obtained by solving Equations 14–16 numerically, to get an idea of what type of gamma-ray signal will be emitted due to the production of runaway electrons. Thus, in this paper, we regard the n_{R} curves as "light curves."

In the n_{R} curves, we distinguish between two main types of signals, namely flashes and glows. Here, flashes are defined as strong and short-lived pulses, coming either alone, like single-pulse TGFs, or in series, like FGFs or multi-pulse TGFs. Glows, on the other hand, are weaker but long-lasting signals, such as GRGs and GGBs. Since we cover the whole possible ranges of durations and intensities, this distinction is somewhat a matter of convention, and we have here tried to choose reasonable values in compliance with the gamma-ray signal classification of Østgaard et al. (2024). To give a demonstration of how we recognised the different signals simulated with our model, Figure 2 shows three examples of solutions resulting from different input values for L_{cloud}, k_{E}, and τ, given $E_0 = 1.04E_{\text{t}}$. For each case, n_{R} is plotted in the top panel and the (total) electric field strength E of the high-field region in the bottom panel.

For each simulation, we located flashes by finding any measurable pulses in the n_{R} curves with durations shorter than 50 ms. By eliminating the flash pulses from the n_{R} curve (through linear interpolation between local minima), we obtained what we shall call a "glow curve," denoted as $n_{\text{R}}^{\text{glow}}$ and plotted as a red dotted curve in panel (b). The

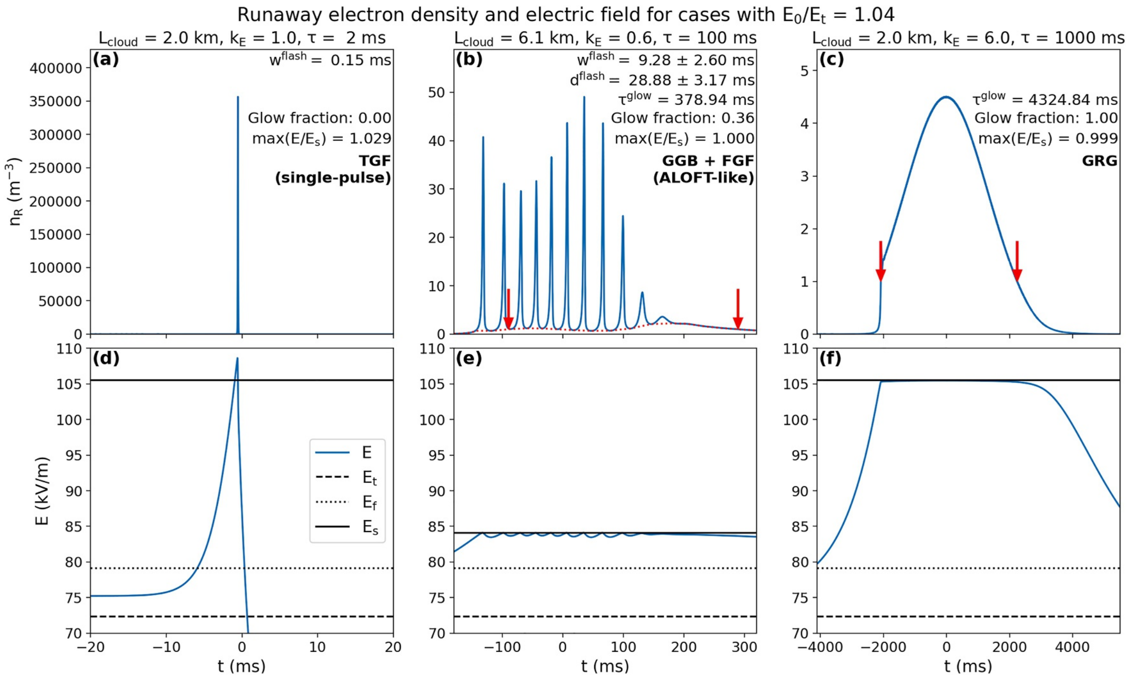

Figure 2. Three different types of signals simulated with our model, resembling a TGF (left), GGB + FGF (center), and GRG (right). Top: number density n_{R} of runaway electrons as function of time. The dotted curve (middle case only) plots the glow component $n_{\mathrm{R}}^{\mathrm{glow}}$. Red arrows mark the start and end of a glow. Bottom: electric field strength E of the high-field region, plotted together with the thresholds E_{t}, E_{f}, and E_{s}. For each simulation, we have printed the following quantities in the top panels: the duration of flash pulses w^{flash}, the separation interval between flash pulses d^{flash}, the duration of the glow (if any) τ^{glow}, the glow fraction $\int_{t_1}^{t_2} n_{\mathrm{R}}^{\mathrm{glow}}\mathrm{d}t / \int_{t_1}^{t_2} n_{\mathrm{R}}\mathrm{d}t$, the electric field amplitude over feedback threshold ratio $\max(E/E_{\mathrm{s}})$, and the classification of the simulated gamma-ray signal.

glow curve is not plotted in panel (a), where $n_{\mathrm{R}}^{\mathrm{glow}} = 0$, nor in panel (c), where $n_{\mathrm{R}}^{\mathrm{glow}} = n_{\mathrm{R}}$. Any parts of the curve where $n_{\mathrm{R}}^{\mathrm{glow}} > 1\ \mathrm{m}^{-3}$ for at least 50 ms were characterized as glows. The start and end points of any glow are marked by red arrows in the plots, where the glow duration τ^{glow}, as printed in panels (b, c), is the time interval between those points. The durations $\left(w^{\mathrm{flash}}\right)$ of the flash pulses, for which the average value (± standard deviation) is printed in panel (b), were recorded as the lengths of the time intervals where, for each pulse, the value of $n_{\mathrm{R}} - n_{\mathrm{R}}^{\mathrm{glow}}$ exceeds 5% of the corresponding peak value of for the given pulse. Knowing the location of each flash pulse, we also measured the separation intervals $\left(d^{\mathrm{flash}}\right)$ between the pulses, with average value and standard deviation also printed in panel (b). For each simulation (in the top panels), we have also printed the glow fraction $\int_{t_1}^{t_2} n_{\mathrm{R}}^{\mathrm{glow}}\mathrm{d}t / \int_{t_1}^{t_2} n_{\mathrm{R}}\mathrm{d}t$, the electric field amplitude over feedback threshold ratio $\max(E/E_{\mathrm{s}})$, and the classification of the simulated gamma-ray signal.

The first case in Figure 2 is an example of a reproduction of a strong single-peak TGF, with all its typical characteristics: the signal has a high peak amplitude $\left(n_{\mathrm{R}} > 3.5 \times 10^5\ \mathrm{m}^{-3}\right)$ but a very short duration $\left(w^{\mathrm{flash}} \sim 150\ \mu\mathrm{s}\right)$, and E goes relatively far beyond the self-sustained feedback threshold E_{s} (by ~ 3 %) right before the strong increase in n_{R} which, through a series of intermediate events (involving ions), ultimately leads to the abrupt discharging of the field. Despite the fact that most observed TGFs are associated with leader discharges, theory shows that gamma-ray flashes similar to TGFs are also fully reproducible with pure feedback discharges (Dwyer, 2012; Liu & Dwyer, 2013), and since the signal reproduced in this case indeed has a shape that matches observed TGFs, we choose to classify it as a TGF. This TGF was simulated with a relatively small high-field region ($L_{\mathrm{cloud}} = 2.0$ km) charged moderately ($k_{\mathrm{E}} = 1.0$) over a short characteristic time scale ($\tau = 2$ ms). From Equation 13, we estimate that the (minimum) charge moment change due to the external force lies around 6 Ckm, which is relatively small.

The second case represents an FGF on top of a GGB. Consisting of 10–11 pulses with intensities several orders of magnitude lower than for typical (strong) TGFs, durations of $\sim$9 ms on average, and an average separation interval $\left(d^{\text{flash}}\right)$ of about 28 ms, the FGF is quite similar in shape to those observed by ALOFT. Thus, we characterized this FGF as "ALOFT-like," despite the fact that in this simulated case, the GGB comes near the end of the FGF, in contrary to most ALOFT observations, where the GGB precedes the FGF. The duration of the glow $\left(\tau^{\text{glow}}\right)$ is about 380 ms (thus classified as a GGB, as we use GRG only for glows that last longer than 1 s). Since the peak intensities of the strongest flashes are more than an order of magnitude above the peak of the glow, the signal might be observed as a pure FGF. In this case, E barely exceeds E_{s} at the peak of each pulse, and after the last pulse, the field is only mildly discharged compared to the TGF case. The FGF was simulated with a quite large high-field region ($L_{\text{cloud}} = 6.1$ km) charged weakly ($k_{\text{E}} = 0.6$) over a moderate characteristic time scale ($\tau = 100$ ms). The charge moment change for this FGF lies roughly around 100 Ckm, which is considerably larger than that for the TGF. This does not contradict the fact that we observe no radio signals together with FGFs, in contrast to TGFs, because the charges move at a considerably slower pace in the FGF case than in the TGF case. In fact, the radio amplitude of a TGF is estimated to be linearly proportional to the time derivative of the current moment or approximately proportional to the inverse square of the TGF duration (Dwyer & Cummer, 2013). Hence, if FGFs have any associated radio emission that scales similarly, one can easily estimate that their radio amplitudes are several orders of magnitude lower than for TGFs and hence below the detectable limit. In terms of pulsations, this FGF is similar in shape to those reproduced with the particle-in-cell (PIC) and Monte Carlo (MC) simulations of Dwyer (2012, 2025), except that in the 2025 paper, the simulation reproduced a GGB-like feature occurring before the FGF, akin to observations.

The latter case represents a GRG with a relatively low intensity but lasting for more than 4 s. Self-sustained feedback is never achieved in this case, as the amplitude of E lies barely below E_{s}. Most of our simulated GRGs (like this one) have a nearly Gaussian shape, which is due to the shape of the external electric field source applied in our cases, despite the fact that most observed GRGs have more arbitrary shapes, though some nearly Gaussian GRGs occasionally occur, as observed with the In-Flight Lightning Damage Assessment System (ILDAS, see Figure 3 of Kochkin et al., 2017). The GRG plotted in this figure was simulated with a high-field region of the same size as the TGF, charged strongly ($k_{\text{E}} = 6.0$) on a long characteristic time scale ($\tau = 1000$ ms). The charge moment change here is about 30 Ckm. One may note that the intensity ratio between the TGF, FGF, and GRG seen in this figure is consistent with the values seen in Table 1 of Østgaard et al. (2024).

The current J_{ext} required to set up the external electric field E_{ext} can be estimated (from displacement current) as $|J_{\text{ext}}| = \epsilon_0 |\text{d}E_{\text{ext}}/\text{d}t| \approx \epsilon_0 k_{\text{E}} E_{\text{f}}/4\tau$ (since the sigmoid function for E_{ext} rises roughly from $k_{\text{E}}E_{\text{f}}/4$ at $t = -\tau$ to $3k_{\text{E}}E_{\text{f}}/4$ at $t = \tau$), which gives $|J_{\text{ext}}| \sim 100\ \mu\text{A/m}^2$ for the TGF case in Figure 2a. That is four orders of magnitude higher than typical current densities in highly convective clouds ($10\ \text{nA/m}^2$ as given by Rakov & Uman, 2003) and most likely only attainable in the presence of a leader or streamer. Thus, in the TGF case, one may assume that there is a leader/streamer that provides the external electric field contribution. For both the FGF and GRG cases in the figure (panels b–c), we find $|J_{\text{ext}}| \sim 1\ \mu\text{A/m}^2$, which is still higher than the typical highly convective currents, although much lower than for the TGF case. Since FGFs are (currently) only seen in large tropical thunderclouds, it is not impossible that these features are produced under convective currents that are significantly stronger than usual. As for the GRG case, it is also possible that convective currents are enhanced during the time the glows are produced.

Though forward-modeling of photon intensities is left out for future work, we may estimate the photon count of our simulated features by assuming that roughly 0.1 photon per electron is emitted from the high-field region (Skeltved et al., 2014). If we assume the horizontal extent of the high-field region to be disk-shaped with a radius $R \sim 5L_{\text{cloud}}$ (keeping the $R \gg L_{\text{cloud}}$ assumption valid), the maximum flux of photons exiting the high-field region is $\max F_\gamma \sim 0.1 \max n_{\text{R}} \pi R^2 c$. By approximating the emitted gamma-ray pulses to be triangularly shaped, we estimate the total photon count of one such pulse (in the case of TGFs and FGFs) to be $I_\gamma \sim \max F_\gamma \cdot 0.5 w^{\text{flash}}$, where w^{flash} is the duration of the pulse. The TGF case of Figure 2a, with $\max n_{\text{R}} = 3.5 \times 10^5\ \text{m}^{-3}$, $w^{\text{flash}} = 150\ \mu\text{s}$, and $L_{\text{cloud}} = 2$ km, is then found to have a maximum photon flux of $\max F_\gamma \sim 10^{21}\ \text{s}^{-1}$ and a total photon count of $I_\gamma \sim 10^{16}$, that is, apparently just slightly fainter than those typically seen with gamma-ray-observing space satellites such as the Atmosphere-Space Interactions Monitor (ASIM, Østgaard et al., 2019). For the strongest pulse of our simulated FGF case (panel b) $\max n_{\text{R}} = 50\ \text{m}^{-3}$, $w^{\text{flash}} \sim 10$ ms and $L_{\text{cloud}} = 6.1$ km (and $R = 30$ km, which is large, though still achievable for tropical thunderclouds), we find $\max F_\gamma \sim 10^{18}$ s and $I_\gamma = 10^{16}$. Thus, even though the photon count of the strongest FGF pulse is of same order of magnitude as the

TGF case, it is spread over a much larger time interval, causing the photon flux to be smaller by three orders of magnitude. Therefore, we would not expect FGFs like this one to be detectable with satellites like ASIM. However, it is here worth mentioning that some of the multi-pulse TGF-like features recorded with Burst and Transient Source Experiment at the Compton Gamma Ray Observatory (CGRO-BATSE, Fishman et al., 1994) have in hindsight been found to be strongly resemblant of ALOFT's FGFs (Sarria et al., 2025). FGFs akin to Figure 2b are however more likely to be detectable from a plane passing closely by, given the right equipment and timing, such as the case of ALOFT. For the GRG case (panel c), with $\max n_{\mathrm{R}} \approx 4\ \mathrm{m}^{-3}, \tau^{\mathrm{glow}} \approx 4$ s, and $L_{\mathrm{cloud}} = 2000$ km we similarly estimate $\max F_{\gamma} \sim 10^{16}\ \mathrm{s}^{-1}$ and $I_{\gamma} \sim \max F_{\gamma} \cdot 0.5\tau^{\mathrm{glow}} \sim 10^{16}$, which means that this glow feature is five orders of magnitude fainter (in terms of photon flux) than the TGF and therefore also undetectable from space, but possibly detectable from by-passing airplanes or balloons.

Figure 2 only briefly demonstrates how to reproduce a TGF, FGF (+GGB), and a GRG with our model using a specified set of values for certain input parameters. In fact, signals similar to these may be reproduced for a vast number of different combinations of values for these input parameters. Because of this, we chose to perform a parameter-space exploration to systematically map for which combinations of E_0, τ, k_{E}, and L_{cloud} we may reproduce TGFs, FGFs, GRGs, and GGBs. For example, such a parametric analysis could give us an estimate of the minimum charge layer separation distance needed to reproduce FGFs similar in shape to those observed with ALOFT. In the next sections, we list the key quantities that we measured from each simulation and how we used those key quantities to classify the output signal in each case.

2.2.1. Key Quantities Recorded From Each Simulation

From each simulation that we performed, as part of our parametric study, we analyzed the simulated curves for n_{R} and E, located any flashes, estimated the glow component $n_{\mathrm{R}}^{\mathrm{glow}}$, and recorded the following key quantities (some of which were printed in Figure 2):

- The amplitude of the strongest flash pulse, $\max n_{\mathrm{R}}^{\mathrm{flash}}$;
- The number of flash pulses, N^{flash};
- The mean duration of flash pulses, w^{flash};
- The mean separation interval between flash pulses, d^{flash};
- The "glow amplitude," $\max n_{\mathrm{R}}^{\mathrm{glow}}$;
- The glow duration, τ^{glow}, defined as the length of the time interval where $n_{\mathrm{R}}^{\mathrm{glow}} > 1\ \mathrm{m}^{-3}$;
- The glow fraction, defined as $\int_{t_1}^{t_2} n_{\mathrm{R}}^{\mathrm{glow}} \mathrm{d}t / \int_{t_1}^{t_2} n_{\mathrm{R}} \mathrm{d}t$, a quantity that tells us how dominant the glow component of the curve is (compared to the flashes) in each case, being equal to 1 if the n_{R} curve is a pure glow curve (no flashes), 0 if the curve consists of flashes only (no glows), and between 0 and 1 if the curve has both glows and flashes;
- The electric field amplitude over feedback threshold ratio, $\max(E/E_{\mathrm{s}})$, which exceeds unity if self-sustained feedback is attained at any point.

Concerning our choice of using $n_{\mathrm{R}}^{\mathrm{glow}} > 1\ \mathrm{m}^{-3}$ as a criterion for measuring the glow duration: we cannot prove here that a density slightly above one runaway electron per cubic meter will lead to any measurable gamma-ray emission (we will get a better idea of this when synthesizing gamma-ray intensities in a future paper). But the idea is that we can assume that some kind of glow will occur as long as the runaway electron density stays above a certain threshold value for a sufficiently long time interval.

2.2.2. Classification of Flashes and Glows

For any flash, or series of flashes, found in the simulated n_{R} curves, as demonstrated in Figures 2a and 2b, we applied the following classification.

1. Signals with at least three observable flash pulses (neglecting any pulses too weak to be detected in a real observation) and with $\max(E/E_{\mathrm{s}}) < 1.005$ were classified as FGFs. The $\max(E/E_{\mathrm{s}}) = 1.005$ threshold used to distinguish FGFs from multi-pulse TGFs (as well as distiguishing weak TGFs from strong single- or double-pulse TGFs) was empirically derived and seemed to work well for our analysis, as signals on each side of this borderline were found to be visually different in nature. Firstly, the amplitudes of the signals are significantly larger when feedback plays a dominant role (as is the case for $\max(E/E_{\mathrm{s}}) \geq 1.005$). Secondly, as will be shown in Section 3, we managed to reproduce some FGFs very similar to those observed by ALOFT still with

$\max(E/E_s)$ slightly above unity. In addition, many cases with $\max(E/E_s)$ barely above 1 did indeed have relatively low amplitudes, therefore fitting well into the regime of FGFs (and weak TGFs). This means that feedback may also play a considerable role on FGFs and weak TGFs. Thus, we found it reasonable to place the boundary between these two regimes at a value for $\max(E/E_s)$ slightly above unity (i.e., 1.005).

(a) Furthermore, FGFs with mean pulse duration $\left(w^{\text{flash}}\right)$ and mean pulse separation interval $\left(d^{\text{flash}}\right)$ between 0.4 and 30 ms were classified as "ALOFT-like," being roughly similar in shape to several of those observed by ALOFT (see Figure 2 of Østgaard et al., 2024).

(b) Other FGFs were labeled as "general FGFs," covering any series of flashes that have too short or long pulse durations and separations compared to those seen with ALOFT but are still too weak to be seen as multi-pulse TGFs.

2. Signals with one or two observable flash pulses and $\max(E/E_s) < 1.005$ were classified as "weak TGFs."
3. Signals with at least one flash pulse with $\max(E/E_s) \geq 1.005$ were classified as (strong) TGFs, as the electric field in these cases goes sufficiently beyond the self-sustained feedback threshold to trigger a dramatic increase in the number density of runaway electrons. Furthermore, our TGFs were subdivided into

(a) Single-pulse TGFs $\left(N^{\text{flash}} = 1\right)$, and

(b) Multi-pulse TGFs $\left(N^{\text{flash}} > 1\right)$.

Thus, despite the fact that the TGFs we observe generally come with lightning leaders (or streamers), which we do not model in our cases, we classify the signal as a TGF as long as pulse(s) is similar in shape to a TGF.

Any glow found in the n_R curves was classified as a

1. GRG if $\tau^{\text{glow}} \geq 1$ s (see Table 1 of Østgaard et al., 2024), otherwise
2. GGB if 50 ms $\leq \tau^{\text{glow}} < 1$ s.

If both glows and flashes were registered at the same time, we usually classified the signal according to both the flash type and the glow type. As an example, a combination of a GGB and an FGF would be classified as a GGB + FGF. However, we neglected the flashes in cases with a glow fraction above 0.95. This threshold for neglecting flashes was chosen empirically, as signals with glow fraction between 0.95 and unity typically looked like a glow with minor spikes on top of it and would probably be seen as a pure glow in any real observation. Likewise, we neglected the glow component of signals with a glow fraction below 0.2, as these signals effectively looked almost like pure flashes (in the case of Figure 2b, the glow fraction is 0.36, and even there the glow component is quite small compared to the flash amplitude).

2.3. The Parameters Explored in This Study

Listed below are the parameters we chose to explore and the applied range of input values.

1. For the vertical charge layer separation distance L_{cloud}, we used 240 different values, ranging from 50 m to 12 km with steps of 50 m. This upper boundary corresponds roughly to the vertical extent of the largest cumulonimbus clouds, according to the National Oceanic and Atmospheric Administration (NOAA). Though the high-field region of a thundercloud never gets that large, we wanted to include some extreme cases to check how large range of charge layer separation distances we really needed to include in order to reproduce all of the previously observed gamma-ray emissions. The self-sustained feedback threshold E_s converges (from infinity) toward E_f as L_{cloud} increases. Consequently, self-sustained feedback will happen more easily for the larger high-field regions than for the smaller ones. Therefore, we expect the value of L_{cloud} to have a significant impact on the shape of the emitted signal.
2. For the relative amplitude k_E of the external electric field contribution, we used 200 different values, ranging from 0.03 to 6 with steps of 0.03. As discussed in the beginning of this chapter, this chosen range for k_E ensures that the charge moment change in most of our simulations is not too large compared to the strongest charge moment changes observed in terrestrial thunderclouds (maybe except for the cases in the upper ranges of L_{cloud} and k_E combined). One may expect that the value of k_E has a scaling effect on the amplitudes of the resulting n_R curves.
3. For the external electric field increase time scale τ, we used nine different values: 2, 5, 10, 20, 50, 100, 200, 500, and 1000 ms. This parameter is expected to have an effect on the duration of the resulting pulses, thus playing a crucial role on whether the resulting signals will resemble TGFs, FGFs, GRGs, or GGBs.

4. For the initial electric field strength E_0, we used three different values, namely 4%, 8%, and 12% above the RREA threshold E_t. Hence, the first two cases start below the minimum feedback threshold E_f (which lies $\sim 9\,\%$ above E_t), while the latter starts above it, but still below the self-sustained feedback threshold E_s (for any realistic value of L_{cloud}). A higher initial electric field strength allows the total field to quench faster, which expectantly ultimately could affect the shape of the resulting gamma-ray signals.
5. The ion attachment length Λ_{ion}—depending on the liquid water content of the cloud, the number density and convective velocity of cloud droplets, the ambient electric field strength, and the charge per droplet—has an estimated range of $3-20$ m (Dwyer, 2005). We ran several cases with $\Lambda_{ion}=3$ m and $\Lambda_{ion}=20$ m to test how the reproducibility of the different hard radiation signals depended on this parameter. Still, most of our cases were run with $\Lambda_{ion}=3$ m, as a larger variety of different phenomena was found to be more easily reproducible with a lower value of this parameter, as will be shown in Section 3.

3. Results

In the following sections, we use the results of our simulations to study where in parameter space the different types of signals (as seen in the simulated n_R curves) belong. Given a specific value of the initial electric field strength E_0 and the ion attachment length Λ_{ion} (= 3 m if nothing else is specified), our results are shown in the form of parameter-space diagrams that map for which values of τ, k_E, and L_{cloud} the different types of signals are reproduced. We also include some parameter-space diagrams for other relevant quantities, such as the number of flash pulses (for TGFs and FGFs) and the electric field amplitude over feedback threshold ratio $\max(E/E_s)$. Furthermore, n_R curves are shown for selected cases to demonstrate the capability of our model to reproduce many different signals akin to those observed in thunderclouds.

3.1. Simulations With $E_0=1.04E_t$

Figure 2 serves as a brief demonstration of how we located and analyzed flashes and glows and determined whether or not self-sustained feedback is achieved. The main results of our analysis are demonstrated in the parameter-space diagrams. Figure 3 shows a parameter-space diagram for signal classification for the cases with $E_0=1.04E_t$. Each panel maps, for a specific value of τ, the classification of the reproduced signal against L_{cloud} and k_E. One should note that in this diagram, reproductions of pure ALOFT-like FGFs and superpositions of (weak) GGBs and ALOFT-like FGFs were put into a joint group labeled “FGF* (ALOFT-like)” (or “FGF* (A)” for short). We did this both to reduce the number of different colors used in this map as well as to easily high-light the parameter-space areas where any ALOFT-like FGFs are reproduced, whether or not they come together with GGBs (especially as some of the underlying GGBs are relatively weak, so that the phenomenon would in real life be observed as a pure FGF). Similar diagrams are shown for the number of flash pulses N^{flash} in Figure 4 and for the electric field amplitude over feedback threshold ratio $\max(E/E_s)$ in Figure 5. Additional diagrams for the other key quantities listed in Section 2.2.1 are included in the supplementary material (Figures S1–S6 in Supporting Information S1). As discussed in Section 2, even though k_E goes up to 6, meaning that the total external electric field contribution is many times higher than E_s, one should note that the total electric field never gets this high, as seen in Figure 5, which shows that E rarely gets higher than 5% above E_s. The fact that the ALOFT-like FGFs seem to cover a relatively small part of parameter space in these diagrams does not contradict the fact that ALOFT detected a relatively high amount of FGFs (30 FGFs and 100 TGFs), since the total amount of gamma-ray signals produced in a thunderstorm is not necessarily uniformly distributed over this parameter space.

By quickly comparing Figures 3–5, one may see that the TGF and FGF regimes mostly coincide with the blue regimes of Figure 5, meaning that the signals are mainly produced under self-sustained feedback. As an exception, the regime of ALOFT-like FGFs coincides primarily with the white or light-red regimes, which means that $\max(E/E_s)$ is roughly equal to or slightly below unity (in a few cases above). The regimes of pure GRGs or GGBs fall into the (light-red) regimes where $\max(E/E_s)$ lies between 0.98 and right below unity, while the regimes characterized by no observable signals coincide with the (dark-red) regimes where $\max(E/E_s)<0.98$. By comparing the different panels of Figure 4, one may see that a more slowly increasing external electric field allows TGFs and FGFs with a larger number of pulses to be produced. As an example, FGFs with more than 30 pulses are reproducible for $\tau=1000$ ms.

From the three leftmost panels of Figure 3, it is clear that cases with a rapidly increasing electric field typically result in TGFs. Especially, the cases with $\tau=2$ ms exclusively produce TGFs (single- or multi-pulse), except for

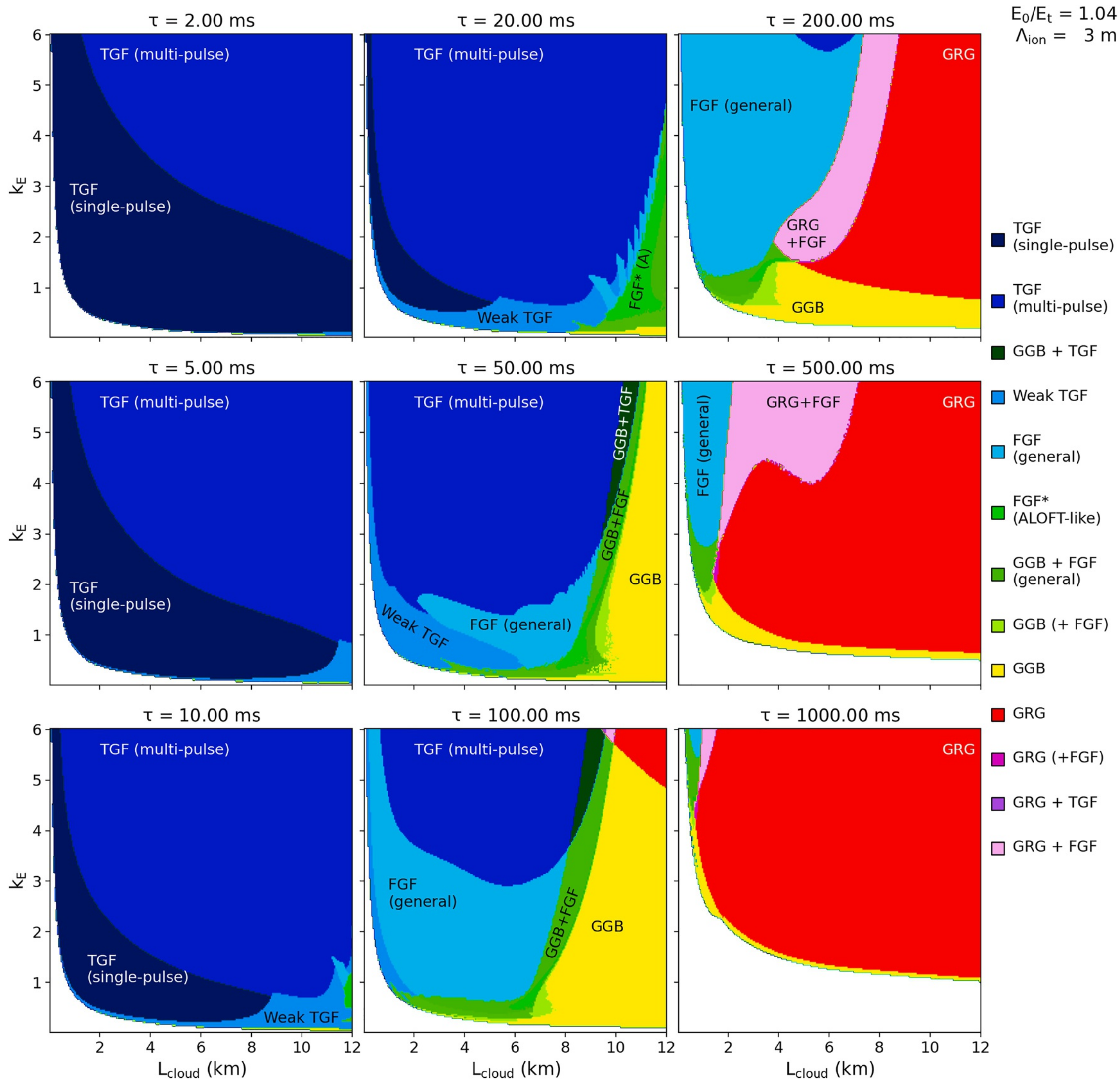

Figure 3. Parameter-space diagram for classification of gamma-ray signals produced with $E_0 = 1.04E_t$. Given a value of τ, each panel maps the signal classification against L_{cloud} and k_E. The "FGF* (ALOFT-like)" grouping, annotated as "FGF* (A)" in the maps, does here cover both pure ALOFT-like FGFs and superpositions of (weak) GGBs and ALOFT-like FGFs.

the very lowest ranges of k_E where no observable signal is produced. Furthermore, we see that multi-pulse TGFs are typically produced for higher values of k_E, while single-pulse for lower values. The minimum value of k_E needed to produce a single-pulse TGF is apparently roughly inversely proportional to L_{cloud}, and the same applies, more or less, to the minimum k_E needed to produce a multi-pulse TGF. For increasing values of τ (up to 20 ms), the boundary between the single- and multi-pulse TGF regimes is shifted downward along the k_E-axis. Hence, multi-pulse TGFs are more easily produced in larger and more slowly charging high-field regions. For $\tau \geq 5$ ms, weak TGFs are produced in the lower ranges of k_E, and for $\tau = 10$ ms, ALOFT-like FGFs are produced in the very highest ranges of L_{cloud} (>11 km).

The $\tau = 20$ ms map has many similarities with the 10 ms one, but the main differences are as follows. ALOFT-like FGFs are reproducible here for values of L_{cloud} down to 10 km. For slightly lower values of L_{cloud} (down to 9 km, given $k_E \approx 1$), other FGFs, not classified as ALOFT-like, are reproduced. This can be anything from FGFs that have a slightly higher mean separation interval between pulses ($d^{flash} > 30$ ms) than those seen with ALOFT to those which lie close to the multi-pulse TGF regime. Within the TGF regimes, as seen in Figure 4, TGFs with a higher number of pulses are produced in this case more easily. For example, with $L_{cloud} \sim 4$ km, the model reproduces TGFs consisting of one to four pulses (given our chosen range of k_E). The boundaries between the

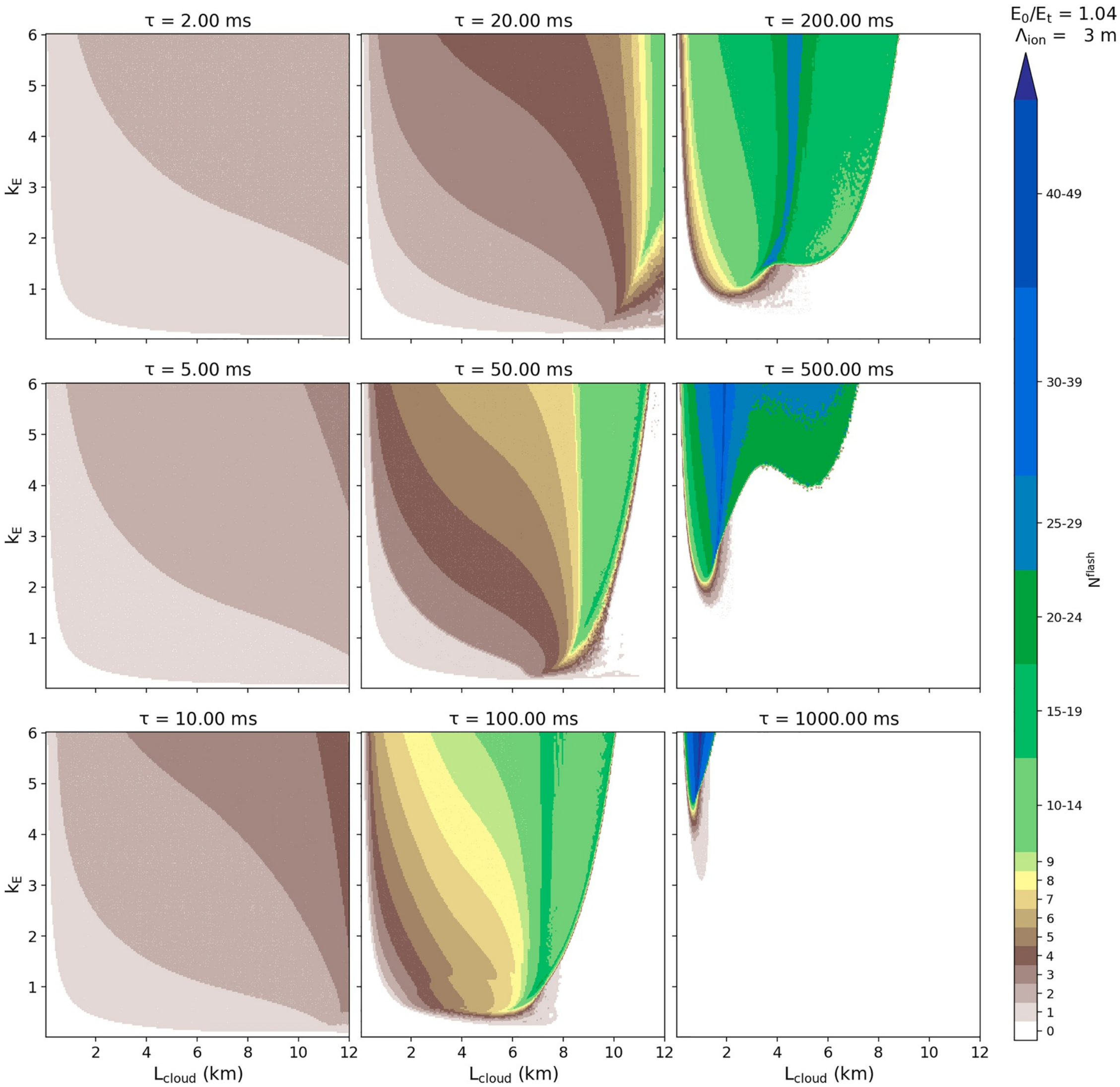

Figure 4. Parameter-space diagram for the number of flash pulses, N^{flash}, in the cases with $E_0 = 1.04E_t$. Layout is the same as for Figure 3.

regimes of different numbers of pulses are also reflected in the diagram for the electric field amplitude over feedback threshold ratio (Figure 5).

Among the cases with $\tau = 50$ ms, some ALOFT-like FGFs are produced for L_{cloud} roughly between 8 and 9 km (for k_{E} between 0.1 and 1). Other FGFs (not ALOFT-like) cover a larger area of the parameter space, reproducible for L_{cloud} between 2 and 9 km, and GGBs are reproducible for $L_{\mathrm{cloud}} \gtrsim 10$ km. Between the TGF/FGF and GGB regimes is a narrow regime of superpositions of GGBs and TGFs/FGFs. No (strong) single-pulse TGFs are reproduced for $\tau \geq 50$ ms, but multi-pulse TGFs with vastly different numbers of pulses (up to ~ 15 pulses as seen in Figure 4) occur. For $\tau = 100$ ms, the boundary between the TGF (multi-pulse) and FGF regimes lies higher up along the k_{E} axis, and the GGBs are reproducible for values of L_{cloud} down to ~ 7 km. Some ALOFT-like FGFs are reproduced for limited values of k_{E} given $L_{\mathrm{cloud}} \approx 6$ km (including the one shown in the middle column of Figure 2).

In the cases with the longer characteristic electric field increase time scales, GRGs are allowed to evolve. With $\tau = 200$ ms, they are produced for any $k_{\mathrm{E}} \gtrsim 1$ given $L_{\mathrm{cloud}} \gtrsim 8$ km, and are reproducible for L_{cloud} even down to 5–7 km given $k_{\mathrm{E}} \sim 1 - 2$. GGBs are reproducible for any $k_{\mathrm{E}} \lesssim 1$ given $L_{\mathrm{cloud}} \gtrsim 4$ km. FGFs are reproducible for any $k_{\mathrm{E}} \gtrsim 1.5$ given $L_{\mathrm{cloud}} \lesssim 4$ km, however, also reproducible for larger high-field regions given larger values of k_{E}. Between the FGF and GRG regimes lies a narrow regime of superpositions of FGFs and GRGs. With $\tau = 500$ ms,

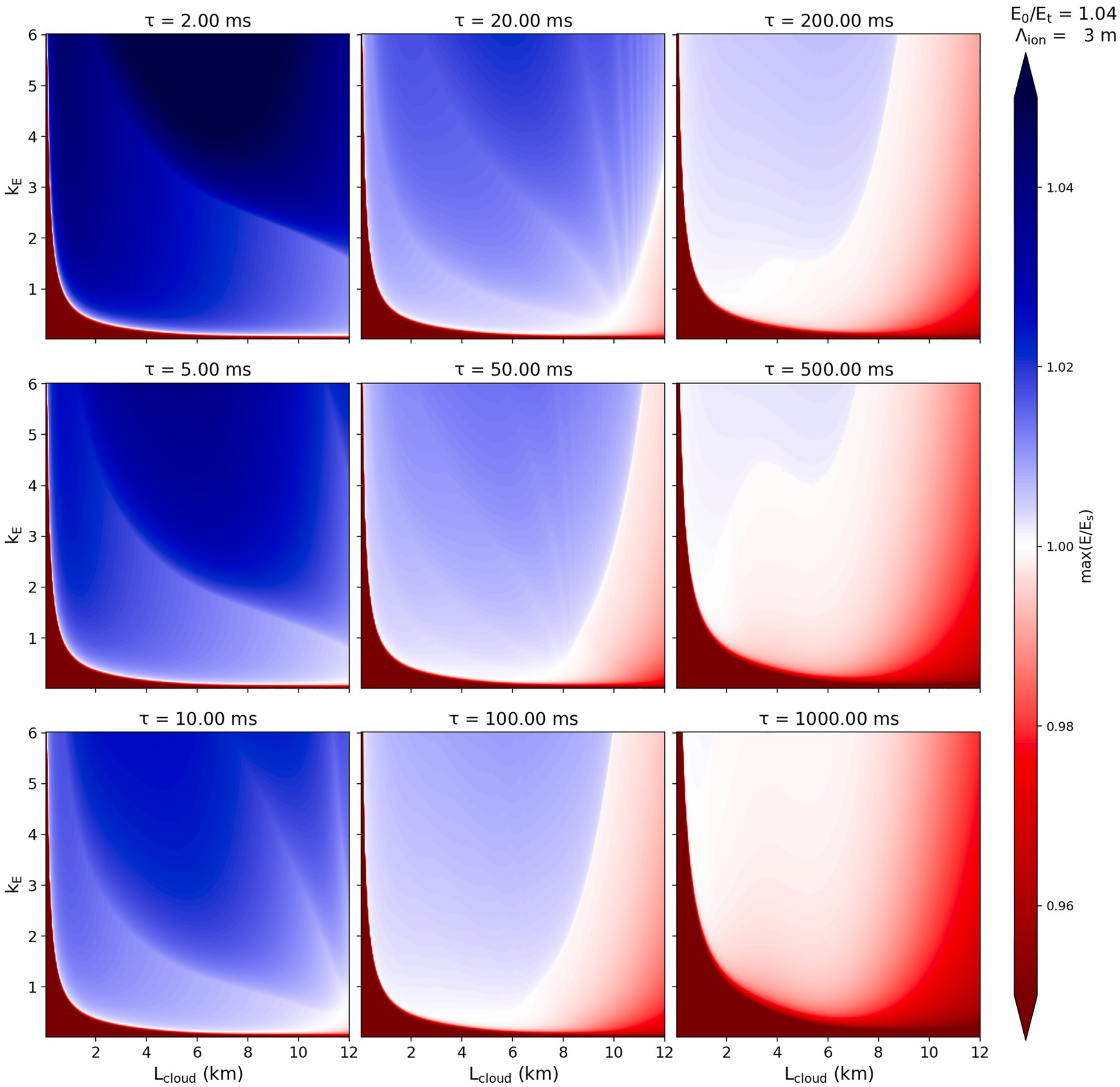

Figure 5. Parameter-space diagram for the electric field amplitude over feedback threshold ratio, $\max(E/E_s)$, in the cases with $E_0 = 1.04E_t$. Layout is the same as for Figure 3.

the GRG regime increases, while the other regimes diminishes (except for GRG + FGF), and with $\tau = 1000$ ms, GRGs dominates the parameter space, being reproduced in most cases except for the lower ranges of k_E where no visible signal is reproduced.

3.2. Simulations With $E_0 = 1.08E_t$

The parameter-space diagram shown in Figure 6 covers the cases where we let the electric field strength start with a value of 8% above the RREA threshold E_t, which is barely below the minimum feedback threshold E_f. This diagram is very similar to the one for the cases with $E_0 = 1.04E_t$ shown in Figure 3. The main difference between these two sets of cases is that most of the boundaries between the different types of signals have been shifted leftward along the L_{cloud} axis as a result of a stronger initial electric field. This is especially evident for ALOFT-like FGFs, which are here reproducible for (vertical) high-field region sizes down to 9 km for $\tau = 20$ ms, down to 7 km for $\tau = 50$ ms, and even down to 5 km for $\tau = 100$ ms, given appropriate values for k_E. The lowest L_{cloud} ranges of the parameter-space diagram are less affected by the increase in E_0, since E_s is still relatively high for these cases compared to E_0. In addition, the maps for the highest ranges of τ are less affected by the increase in E_0, except for the very highest ranges of L_{cloud} (where E_s is closer to E_0), because the external electric field increase time scale is comparable to the relaxation time of the cloud.

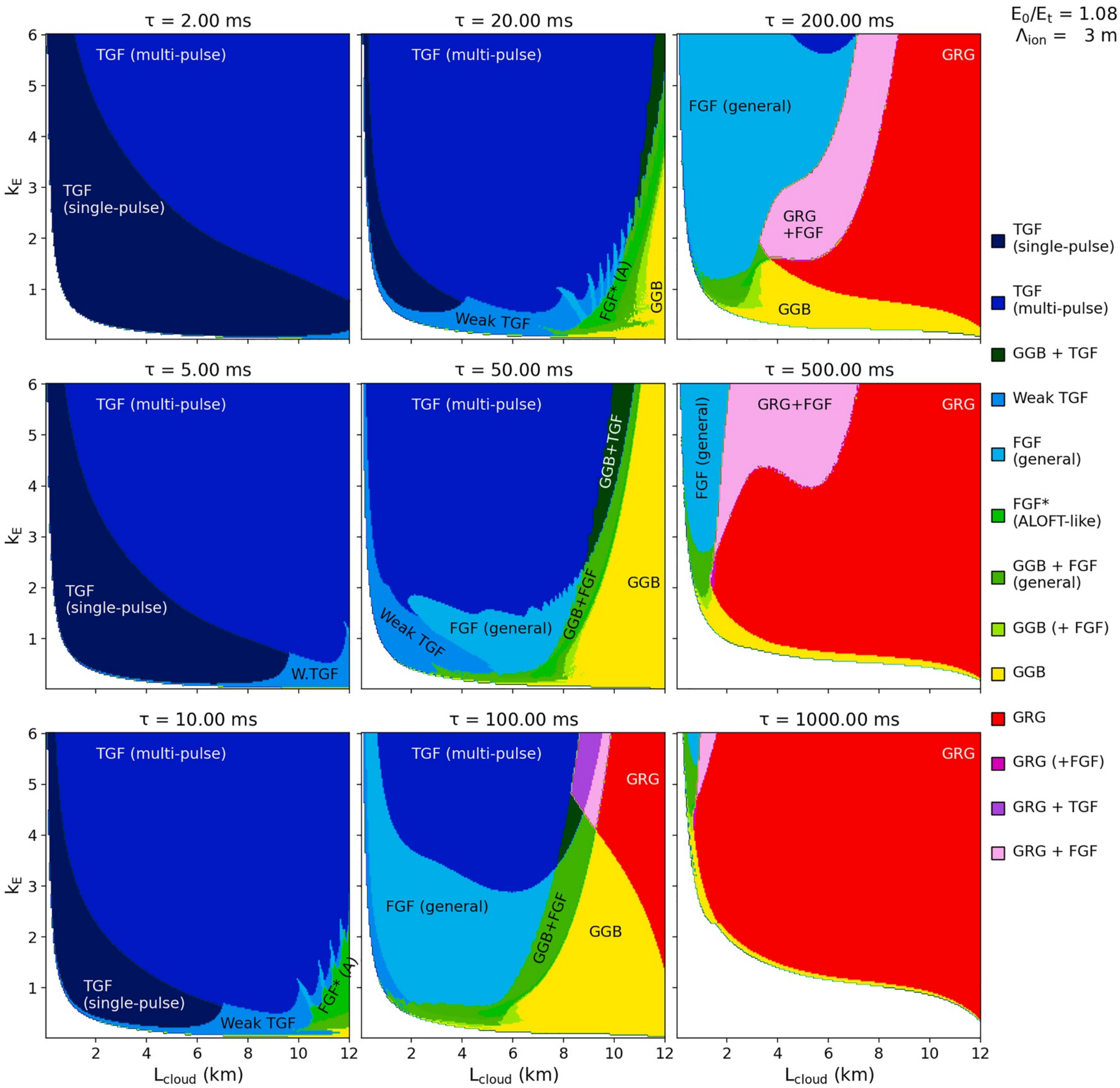

Figure 6. Parameter-space diagram for classification of gamma-ray signals produced with $E_0 = 1.08 E_t$. Layout is the same as for Figure 3.

Figure 7 plots the n_R curves of nine selected cases with $E_0 = 1.08 E_t$ and $L_{cloud} = 5.30$ km. The top three cases ($\tau = 20$ ms) are all reproductions of (weak and strong) TGFs. As expected, the increase in the value of k_E leads to a stronger amplitude. Among the three middle cases ($\tau = 50$ ms), the first case ($k_E = 0.5$) reproduces a weak (double-pulse) TGF followed by a very weak (possibly not observable) GGB. The other two middle-panel cases are both reproductions of FGFs, though with considerably larger separation intervals between the pulses compared to those seen with ALOFT. The first bottom-panel ($\tau = 100$ ms) case is apparently a GGB topped by a weak TGF. In a real observation, this would probably be seen as a pure GGB. The second one represents a typical FGF, very much like those seen with ALOFT, superposed with a (rather weak) GGB. The latter is another FGF with a relatively large mean separation interval between pulses, hence not an ALOFT-like FGF. The charge moment change of the cases shown in this figure are estimated to range from about 20 to 100 Ckm, all within physically reasonable limits. It is here worth mentioning that the TGFs and FGFs observed with ALOFT at 29.7.2023 (Østgaard et al., 2024), several of which are similar to those shown in this figure, were apparently produced in a high-field region with a vertical extent of $\sim$5 km (Marisaldi et al., 2024).

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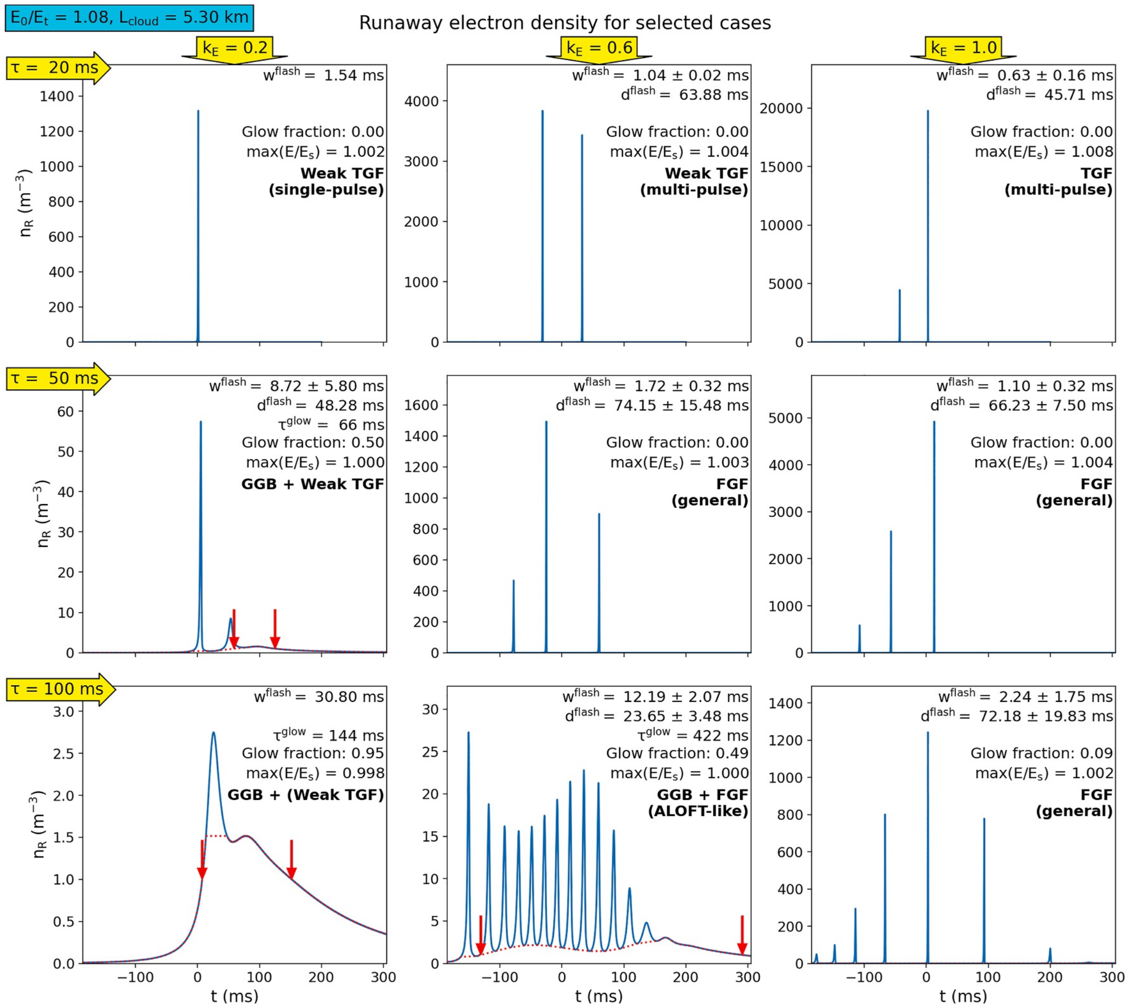

Figure 7. Number density n_R from selected cases with $E_0 = 1.08E_t$. Layout is the same as in Figures 2a–2c.

3.3. Simulations With $E_0 = 1.12E_t$

The parameter-space diagram for the cases with $E_0 = 1.12E_t$ (i.e., well above E_f, though still below E_s for the chosen range of L_{cloud}) is included in the Figure S7 of Supporting Information S1. Apart from some unpredictable behavior in the cases with $L_{cloud} > 8$ km (due to an unstable background situation, $E_0 \sim E_s$), the effect of going from $E_0 = 1.08E_t$ to $E_0 = 1.12E_t$ is mainly that the boundaries between the different regimes (except for the lowest ranges of L_{cloud} and the highest ranges of τ) are shifted further to the left along the L_{cloud}-axis, similar to what we saw when going from $E_0 = 1.04E_t$ to $E_0 = 1.08E_t$. Consequently, FGFs similar to those seen with ALOFT are reproducible here for values of L_{cloud} down to $\sim$7 km for $\tau = 20$ ms, for $L_{cloud} \sim 6$ km for $\tau = 50$ ms, and for L_{cloud} almost down to 4 km in the cases with $\tau = 100$ ms, given appropriate values for k_E. As an example, n_R curves similar to those produced with $E_0 = 1.08E_t$ and $L_{cloud} = 5.3$ km, as shown in Figure 7, could be reproduced with $E_0 = 1.12E_t$ given only $L_{cloud} = 4.1$ km, which are shown in the Figure S8 of Supporting Information S1.

3.4. Cases With Increased Ion Attachment Length

Parameter-space diagram for the $E_0 = 1.08E_t$ cases with $\Lambda_{ion} = 20$ m is shown in Figure 8. By comparing this with the corresponding results for $\Lambda_{ion} = 3$ m seen in Figure 6, we see that the main consequence of the increased ion attachment length is that most of the parameter-space boundaries between the regimes for the different signal types have been shifted upward along the k_E-axis. This is basically due to the fact that these cases with a higher ion free path require a stronger electric field to reproduce the same number of runaway electrons, because a larger

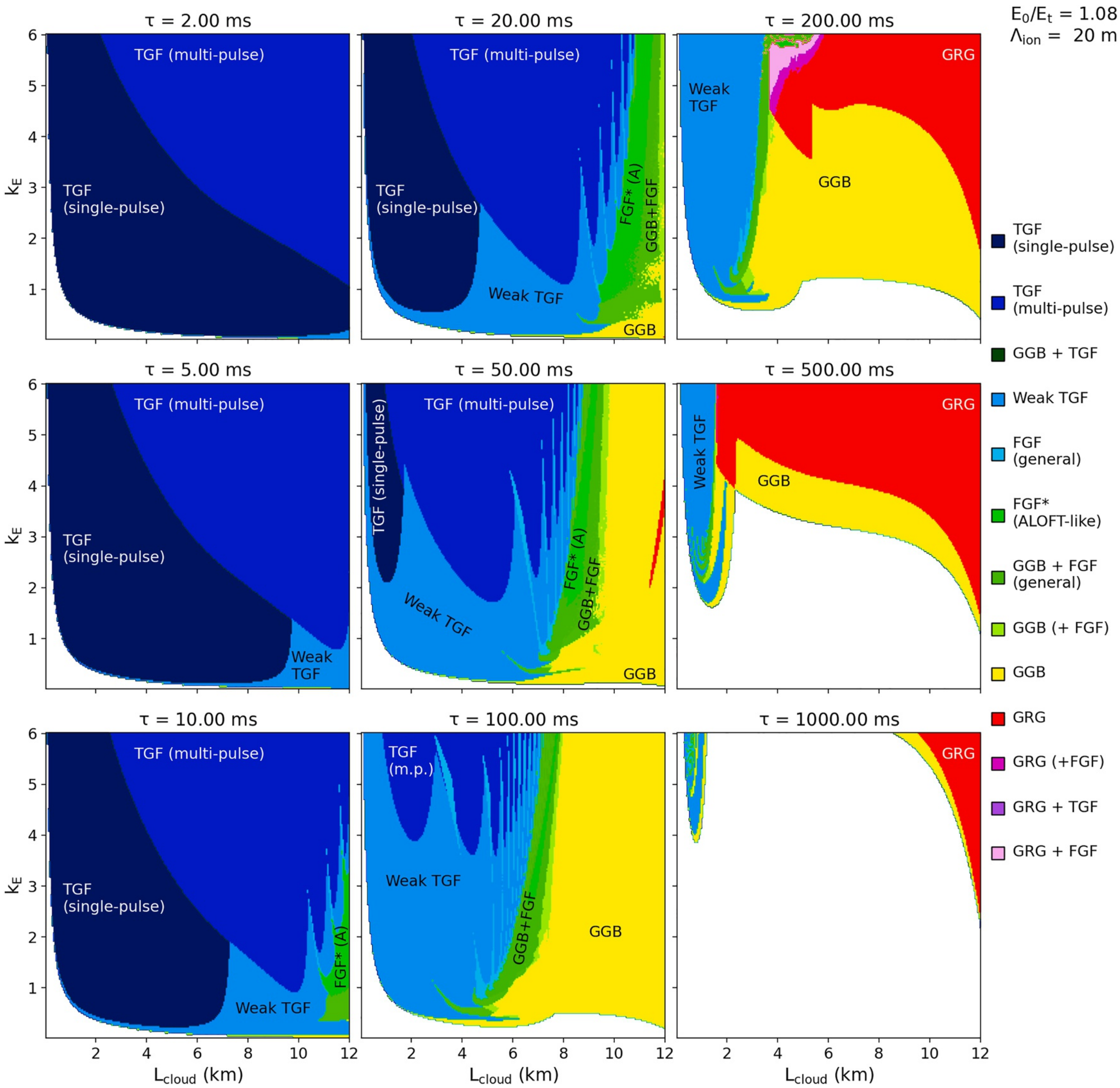

Figure 8. Parameter-space diagram for classification of gamma-ray signals produced with $E_0 = 1.08E_t$ and $\Lambda_{ion} = 20$ m. Layout is the same as for Figure 3. The huge white regions seen in the rightmost panels correspond to cases where n_R never gets sufficiently high to produce detectable gamma-ray signals.

attachment length leads to a higher discharging current, for which one needs to compensate by enhancing the external field source. Especially, we see that a stronger external electric field contribution is needed in order to produce multi-pulse TGFs. We also see that, for τ between 10 and 100 ms, FGFs and weak TGFs are reproduced for a wider range of k_E values. In the higher ranges of τ, GRGs are also more difficult to produce, and with $\tau = 1000$ ms, no observable signal is produced unless the cloud undergoes an extremely high increase in the external field (which would correspond to an unrealistically high charge moment change). Apart from those differences, these cases with increased ion free path also reproduce the whole range of observable gamma-ray signals associated with thunderstorms.

4. Discussion

In this paper, we have developed a new model for reproduction of the different types of gamma-ray emission observed in thunderstorms, based on the RFD theory of Dwyer (2007), by solving a set of ODEs that resemble a Lotka-Volterra predator-prey model. This is a very different approach from previous RFD-based models, such as Dwyer's models, which apply PIC and MC simulations. One great advantage with our model is the simplicity and consequently short calculation time which has allowed us to perform a deep parametric exploration. Using the curves for the resulting runaway electron number density n_R as a proxy for the light curves of the bremsstrahlung

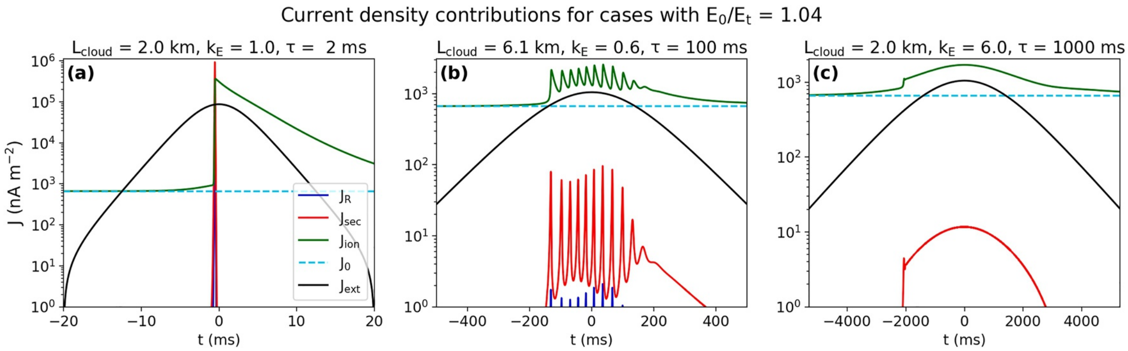

Figure 9. Current density distribution for the signals from Figure 2, that is, a TGF (left), an FGF (middle), and a GRG (right). Plotted in each panel are the current densities of RRE (J_R, blue curve), secondary electrons (J_{sec}, red curve), and ions (J_{ion}, green curve) along with the background current J_0 (cyan dotted curve) needed to maintain the background electric field E_0 and the current J_{ext} (black curve) required to set up the electric field E_{ext}.

emitted from the runaway electrons, our results show that our model is capable of simulating signals which are similar in shape to both TGFs, FGFs, GGBs, and GRGs, depending on the vertical charge layer separation distance, the characteristic time scale and amplitude of the external electric field contribution, and the initial electric field strength.

One major discrepancy between our modeled TGFs and their observational counterparts is the absence of leader (or streamer) discharges, where our TGFs are produced purely by feedback discharges, despite most observations indicating leader (or streamer) discharges being present in the production of TGFs. However, we do not disprove the role of leaders/streamers in the production of TGFs by showing that they can be reproduced by feedback discharges. Neither does the observed presence of leaders rule out the possibility of RFD playing a role in TGFs. They could possibly be the result of leader discharges enhanced by feedback discharges. Also, our model does not say anything about where the external electric field component, which acts as a driver in all hard radiation signals mimicked in our model, comes from. In the TGF cases, the modeled external electric field contributions may in fact be interpreted as contributions from leaders/streamers. Other notable discrepancies between our model and observations are the nearly Gaussian shape of our reproduced GRGs, where most observed GRGs have relatively arbitrary shapes, and that our modeled FGFs lack the long pulses at the beginning of the train as observed with ALOFT and reproduced by Dwyer (2025). This has to do with the fact that we applied an ad hoc external field source in our model, where the field strength as function of time was in all our cases a sigmoid function (where we only varied the amplitude and temporal width). Nevertheless, our model allows for any kind of (continuous) function for this external field, which may be explored more in future experiments.

We also found that our simulated features are driven by relatively large external currents, given the fact that convective cloud currents are typically of order 10 nA m^{-2}, and this deviation may be improved upon when expanding to 1D as well. Figure 9 displays the current density J_{ext} needed to set up the external electric field E_{ext} used to generate the features shown in Figure 2 plotted together with the current density of RRE (J_R), secondary electrons (J_{sec}), and ions (J_{ion}) along with the current J_0 needed to maintain the background electric field E_0. For the TGF case (left panel), it is clear from the black curve that a leader/streamer is indeed needed to generate the current required for setting up the external electric field that drives this feature. From the latter two panels, we see that the current related to the external field that drives the simulated FGF and GRG is significantly weaker, getting only one to two orders of magnitude above the typical values for convective cloud currents, but only for a time interval of a few seconds (<1 s for the FGF case), hence still fairly within reasonable limits for the assumption that FGFs and GRGs could be driven by RFD caused by electric fields set up by convective cloud motion. However, as the current needed to induce the external field in our simulations scale with k_E, it is likely that the realistic domain of our parameter-space diagrams (Figures 3–6 and 8) is limited to the lower half (or less) of each panel.

Our parametric study shows the following distribution of TGFs, GRGs, FGFs, and GGBs in our chosen parameter space. In one end, our model indicates that TGFs (if generated by feedback discharges) are mainly produced when the electric field in the high-field region of the cloud increases on a relatively short characteristic time scale ($\tau \lesssim 20$ ms). This is because the electric field in fact increases so fast that it exceeds far beyond the threshold for self-sustained feedback, hence leading to a rapid increase in the number of runaway electrons according to the RFD mechanism, before the self-quenching effects starts to dominate, causing the electric field to decline quickly. We have also seen that larger high-field regions and stronger external electric field contributions allow for multi-pulse TGFs to occur. TGFs can also be produced under more moderate electric field increase time scales (∼50–100 ms), given a sufficiently strong external field amplitude and a not too large high-field region.

In the other end, the model reproduces the GRGs when the electric field is allowed to build up over sufficiently long characteristic time scales ($\tau \gtrsim 200$ ms), which causes the field to eventually discharge even before it exceeds the self-sustained feedback threshold (hence avoiding the self-quenching effects due to feedback). However, feedback is present during GRGs, though it is not self-sustained, as the peak value of the total electric field typically lies $1 - 2$ % below the self-sustained feedback threshold. For external electric field increase time scales longer than one second, GRGs are produced almost exclusively. Both TGFs and GRGs can be produced from relatively small high-field regions (for L_{cloud} as low as $1 - 2$ km in the case of GRGs and for even smaller L_{cloud} in the case of TGFs).

In the middle ground, FGFs and GGBs are indicatively produced. Especially FGFs that are similar in shape to those seen with ALOFT (except for the lack of long pulses at the beginning), with typical pulse durations of a similar order of magnitude as the separation interval between pulses, indeed represent a missing link between TGFs and GRGs, as demonstrated by Østgaard et al. (2024). This is very evident in the fact that the associated electric field strength gets very close to, or barely exceeds, the self-sustained feedback threshold as the FGF is produced, instead of remaining below it (in the case for the GRGs) or going far beyond it (in the case for the strong TGFs). In general, our model indicates that ALOFT-like FGFs require a relatively large high-field region to be produced, which may explain why they currently have only been observed in large tropical thunderclouds (Østgaard et al., 2024). They are best reproduced with moderate external electric field increase time scales ($\tau \sim 50 - 100$ ms). With shorter time scales ($\tau \lesssim 20$ ms), these FGFs require a very large, possibly unrealistic, charge layer separation distance ($L_{\text{cloud}} \gtrsim 10$ km) to be produced.

Our model shows that the minimum charge layer separation distance required to reproduce ALOFT-like FGFs decreases with higher values of the initial electric field strength. Given $\tau = 100$ ms, they can be reproduced for $L_{\text{cloud}} \sim 6$ km with $E_0 = 1.04E_{\text{t}}$, $L_{\text{cloud}} \sim 5$ km with $E_0 = 1.08E_{\text{t}}$, and $L_{\text{cloud}} \sim 4$ km with $E_0 = 1.12E_{\text{t}}$. In fact, we did some quick attempts to reproduce these FGFs for initial field strengths below the RREA threshold, but found that they would require an unrealistically large charge layer separation distance ($L_{\text{cloud}} > 10$ km) to be reproduced, which also would imply unrealistically large charge moment changes (much larger than what is observed in the strongest sprite-producing lightnings seen on Earth). Because of this, we do not include any parameter-space diagram for cases with $E_0 < E_{\text{t}}$ in this paper. In reality, the electric field inside the cloud is initially far below the RREA threshold before starting to charge. A possible scenario could be that the cloud starts with a relatively weak internal electric field which builds up very slowly (on time scales of several seconds or minutes) until reaching a magnitude close to the minimum feedback threshold (as in the cases of $E_0 = 1.08E_{\text{t}}$ and $E_0 = 1.12E_{\text{t}}$) before an external source kicks in, causing the electric field to grow on a more moderate time scale (∼100 ms) until self-sustained feedback is barely attained, ultimately leading to the production of FGFs.

The results of our parameter-space exploration also indicate that it is theoretically possible to produce a wide range of FGFs that are not similar in shape to those seen with ALOFT, as seen especially in Figures 3 and 6 (center and right columns). After all, FGFs are much weaker than TGFs and therefore rarely detected from space, except for two that has been detected by ASIM (Østgaard et al., 2024) along with those that were recorded with BATSE but mislabeled as multi-pulse TGFs. Thus, for all we know, there could exist a wide range of different FGFs that are emitted from thunderclouds, of which only a small fraction has been recorded. It is also possible that several of those (general) FGFs simulated in our model have been detected but rather been confused with (weak) multi-pulse TGFs, or the other way around (i.e., some of our modeled "general FGFs" could possibly in fact be multi-pulse TGFs). Observations, however, indicate one clear difference between FGFs and multi-pulse TGFs. In contrast to (single- and multi-pulse) TGFs, FGFs do not coincide with radio or optical emissions, although some of them are followed by narrow bipolar events (NBEs, Østgaard et al., 2024). We have not yet synthesized these types of emission with our model, and therefore we had to apply some alternative criteria for distinguishing our FGFs from multi-pulse TGFs. It is also possible that any FGFs may have some associated radio emission, just too weak to be detected, since the radio amplitude is estimated to scale inversely with the square of the pulse durations, and FGF pulses have both longer durations and lower intensities than TGF pulses. Hence, the boundary between FGF and (weak) multi-pulse TGF regimes could be slightly more unclear than currently thought. The purpose of this study, however, was not to draw clear borders between the different regimes, but rather to demonstrate the capability of models based on RFD theory in reproducing a wide range of different gamma-ray emissions detected in thunderstorms and to get an idea of where in parameter space the different emission types are located relative to each other, which we indeed have done successfully.

As our 0.5D model neglects variation along the vertical axis in any of the number densities as well as the electric field, we acknowledge that our results give only rough estimates for where in parameter space the different types of signals are produced. Another error bar is that we assumed all charges to be concentrated in two layers of opposite polarity. In any real cloud, the charge distribution along the vertical axis goes more gradually from positive to negative polarity. When we take this into account in future simulations, we might see a notable shift in the parameter-space boundaries between the regimes of TGFs, FGFs, GRGs, and GGBs. Especially, TGFs are probably produced locally ahead of leader/streamer fields, hence depending quite a lot on the local conditions, so the reproduction of these features could change significantly when going from 0.5D to 1D. Furthermore, as our model assumes a high-field region which is much larger horizontally than vertically, we did not here explore the effects of a finite transverse size of the high-field region (Sedelnikov et al., 2025). Cases where the high-field region is smaller horizontally than vertically, such as for leaders, could yield notably different results and should be delved into in future works.

Our parameter-space exploration includes charge layer separation distances of up to 12 km, which might seem unrealistically large. Standard models of cumulonimbus clouds (such as those given by NOAA) typically assume a total vertical extent of 10–12 km, where the positive charge layer lies only 4 km above the negative charge layer (Phillips, 1967). Though tropical thunderclouds may achieve charge layer separation distances considerably larger than the average thundercloud, recent research still indicates a possibly upper limit of about 6 km (Marisaldi et al., 2024). This does not mean that the right half of our parameter-space diagrams ($L_{\text{cloud}} > 6$ km) should be disregarded. As discussed above, the parameter-space boundaries could shift notably when expanding the model to 1D, meaning that an FGF reproduced with $L_{\text{cloud}} = 8 - 9$ km in our 0.5D model, for example, might be reproduced with L_{cloud} closer to 6 km in a future 1D simulation. Even if the results for $L_{\text{cloud}} > 10$ km are in the unrealistic regimes, they are still valuable in demonstrating that, for instance, the minimum charge layer separation distance required for reproducing FGFs with short charging times ($\tau \lesssim 20$ ms) is unrealistically high but can be reduced by increasing the charging time ($\tau \sim 50 - 100$ ms) as well as the initial electric field strength $(E_0 \sim E_{\text{f}})$. The most important results of this analysis are not the quantitative ones, but rather the qualitative ones, and indeed, our 0.5D approach has given us a qualitative idea about where in parameter space the different regimes are located relative to each other. A good example here is the aforementioned conclusion that the FGFs observed with ALOFT requires a large charge layer separation distance to be reproduced compared to TGFs and GRGs, in agreement with their current status of appearing exclusively in large tropical thunderclouds. We expect future parametric studies to qualitatively provide similar conclusions even after expanding to 1D, in addition to providing more accurate quantitative results as well.

In most of our simulations, we assumed the ion attachment length Λ_{ion} to be in its lowest possible range (3 m), which means that we assumed a relatively high liquid water content and density of water droplets. We also performed a set of simulations with $\Lambda_{\text{ion}} = 20$ m (Figure 8), where the results showed that most of the different types of hard radiation signals known to be emitted from thunderclouds can also be produced in the upper range of values for Λ_{ion}. However, in that case, some of the signal types, especially GRGs and multi-pulse TGFs, require significantly stronger external electric field contributions to be produced, which also implies a higher charge moment change.

Despite our simplified 0.5D setup with the application of an ad hoc external field source for charging the cloud, this parameter-space exploration has given us a qualitative insight into how the shape of the signals in the curves of the runaway electron density (which serve as a proxy for the emitted gamma-ray signals), as a result of the RFD mechanism, depend on the main charge layer separation distance in the cloud as well as how fast and how strongly the cloud is charged. For future research, the natural next step will be to synthesize gamma-ray emissions as well as radio and optical signals, and possibly NBEs, from our simulated runaway electron densities and compare them with real observations (e.g., from ALOFT or ASIM). This will give an even better idea about the physical conditions required to produce the different types of emission observed in thunderstorms. Especially, this would also help us to distinguish our simulated FGFs from multi-pulse TGFs much more precisely. In order to obtain

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even more accurate reproductions of any kind of emission from thunderclouds, hence also more accurate parameter-space exploration, another necessary improvement is to expand our model to 1D. With this improvement, our model could also let the electric field evolve self-consistently (instead of using an ad hoc external field) by the advection of negative and positive charges inside the cloud.

With the implementation of the above-mentioned future improvements, we expect to get better and more accurate insights into the role that the RFD mechanism may play in several hard radiation phenomena seen in thunderstorms. This could have a considerable impact on lightning studies. As an example, gamma-ray measurements could become a new method of remotely reconstructing large-scale electric field inside thunderstorms. Despite more than a hundred years of research, lightning initiation remains poorly understood as of today, especially because the intense electric fields considered necessary for the initiation exceed those typically measured in thunderclouds. Recent papers, though, claim to have brought us closer to understanding the mechanism behind lightning initiation, involving possible candidates such as cosmic ray showers (Shao et al., 2025) and photoelectric feedback discharges (Pasko et al., 2025). An even deeper understanding of this process could potentially be acquired if classical breakdown models for thunderclouds were to be replaced with models based on RFD theory. Moreover, since it is well known that gamma-rays in the atmosphere contribute to the production of NO_x ions that can harm the ozone layer (Pickering et al., 1998; Pollack et al., 2016; Schumann & Huntrieser, 2007), an improved understanding of the mechanism that produces hard radiation in thunderstorms can also have an important impact on atmospheric chemistry.

Appendix A: The Background Ion Source of the Cloud

In our code, the background ion source of the cloud, which sustains its background conductivity, is given by

$$S_{\mathrm{ion}}^{\mathrm{cloud}} = n_{\mathrm{ion}}^{\mathrm{cloud}} v_{\mathrm{th,ion}}(E_0) - S_{\mathrm{ion,\ R}}^{\mathrm{cloud}}, \tag{A1}$$

where the background thermal ion velocity $v_{\mathrm{th,ion}}(E_0)$ is found by inserting the initial background electric field strength E_0 into Equation 26. The latter term of the above equation (on which the first term depends) denotes the background ion source due to runaway electrons,

$$S_{\mathrm{ion,\ R}}^{\mathrm{cloud}} = n_{\mathrm{R}}^{\mathrm{cloud}} \alpha_{\mathrm{ion}} v_{\mathrm{R}}, \tag{A2}$$

where the background number density of runaway electrons,

$$n_{\mathrm{R}}^{\mathrm{cloud}} = n_{\mathrm{R0}}(E_0), \tag{A3}$$

is found by inserting the initial background electric field strength, E_0, into Equation 25. Finally, the first term of Equation A1 denotes the background ion density,

$$n_{\mathrm{ion}}^{\mathrm{cloud}} = \frac{\sigma_{\mathrm{ion}}^{\mathrm{cloud}}}{2e\mu_{\mathrm{ion}}}, \tag{A4}$$

where the background ion conductivity $\sigma_{\mathrm{ion}}^{\mathrm{cloud}}$ can be found by splitting up the term σ^{cloud} given by Equation 17 into $\sigma^{\mathrm{cloud}} = \sigma_{\mathrm{ion}}^{\mathrm{cloud}} + \sigma_{\mathrm{NR}}^{\mathrm{cloud}} + \sigma_{\mathrm{sec}}^{\mathrm{cloud}}$, so that

$$\sigma_{\mathrm{ion}}^{\mathrm{cloud}} = \sigma^{\mathrm{cloud}} - \sigma_{\mathrm{NR}}^{\mathrm{cloud}} - \sigma_{\mathrm{sec}}^{\mathrm{cloud}}, \tag{A5}$$

with the background conductivity of runaway and secondary electrons respectively given by

$$\sigma_{\mathrm{NR}}^{\mathrm{cloud}} = \frac{n_{\mathrm{R}}^{\mathrm{cloud}} e v_{\mathrm{R}}}{E_0}, \tag{A6}$$

$$\sigma_{\mathrm{sec}}^{\mathrm{cloud}} = n_{\mathrm{sec}}^{\mathrm{cloud}} e\mu_e(E_0), \tag{A7}$$

with $\mu_e(E)$ given by Equation 20. In the latter equation, the background number density of secondary electrons is given by

$$n_{\text{sec}}^{\text{cloud}} = \frac{S_{\text{ion, R}}^{\text{cloud}}}{\nu_{\text{att}}(E_0)}, \tag{A8}$$

with the attachment rate $\nu_{\text{att}}(E)$ given by Equation 24.

Appendix B: Lotka-Volterra Predator-Prey Model

It can be shown that Equations 14–16 are approximately analogous to a Lotka-Volterra predator-prey model. That is a system of equations on the form

$$\frac{\mathrm{d}\tilde{E}}{\mathrm{d}t} = \tilde{\alpha}\tilde{E} - \tilde{\beta}\tilde{E}\tilde{\sigma}, \tag{B1}$$

$$\frac{\mathrm{d}\tilde{\sigma}}{\mathrm{d}t} = -\tilde{\gamma}\tilde{\sigma} + \tilde{\delta}\tilde{E}\tilde{\sigma}, \tag{B2}$$

where the electric field $\tilde{E}$ is the "prey," the conductivity $\tilde{\sigma}$ the "predator," and the coefficients $\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}$, and $\tilde{\delta}$ are all >0. In our case, $\tilde{E} \equiv E_{\text{cloud}}$ and $\tilde{\sigma} \equiv (J_{\text{R}} + J_{\text{sec}} + J_{\text{ion}})/E_{\text{cloud}}$. The terms on the right-hand side may be interpreted as follows:

- $\tilde{\alpha}\tilde{E}$ is the charging current term, which in our cases is constant ($=J_0$), meaning that $\tilde{\alpha} \propto 1/E$ (hence not a perfect Lotka-Volterra system);
- $\tilde{\beta}\tilde{E}\tilde{\sigma}$ is the discharging current term $\left(\tilde{\beta} \equiv 1/\epsilon_0\right)$;
- $\tilde{\gamma}\tilde{\sigma}$ is the ion removal term (in our model, $\tilde{\gamma} \sim \eta_{\text{ion}}$);
- $\tilde{\delta}\tilde{E}\tilde{\sigma}$ is the ion creation term due to RREA (the higher $\tilde{E}$).

Appendix C: Detailed Derivation of Equation 6 for the Feedback Factor γ

Consider development of RREA and RFD in a uniform electric field $\mathbf{E} = -\hat{x}E$. The 1D drift equations for n, the number density of the RRE, and n_{B}, the number density of backward-moving seed producers (energetic photons or positrons), are:

$$\left.\begin{aligned} \partial_t n/v_R + \partial_x n &= Rn + Cn_{\text{B}} \\ \partial_t n_{\text{B}}/v_P - \partial_x n_{\text{B}} &= Bn - An_{\text{B}} \end{aligned}\right\} \tag{C1}$$

where $v_R, v_P \approx c$ are the drift speeds of RRE and seed producers, correspondingly. The coefficients of this system are the spatial rates of the following reactions: $R = \frac{E - E_t}{U_R} = 1/\lambda$—RRE avalanche, C—conversion of seed producers into seeds (as a result of Compton scattering for photons or annihilation for positrons), B—conversion of RRE into seed producers (bremsstrahlung or pair production), and $A = 1/\lambda_a$—seed producer absorption.

Let us solve C1 in a steady-state case ($\partial_t = 0$) by assuming $\propto e^{\kappa x}$ dependence. The characteristic equation is

$$(\kappa - R)(\kappa - A) = -BC$$

This is a quadratic equation for κ which may be solved precisely. However, RFD process is characterized by the weakness of the seed producing mechanism, so we can assume that the right-hand side of this equation, BC, is small. Then we find $\kappa_1 \approx R$ and $\kappa_2 \approx A$, and the general solution of the steady-state system given by Equation C1 is

$$\left.\begin{aligned} n(x) &= c_1 D e^{Rx} + c_2 C e^{Ax} \\ n_{\mathrm{B}}(x) &= -c_1 B e^{Rx} - c_2 D e^{Ax} \end{aligned}\right\}$$

where we introduced a useful value of $D = R - A = 1/\lambda - 1/\lambda_a$ (the "net" rate for avalanching with attenuation of producers). The coefficients $c_{1,2}$ have to be determined from the boundary conditions. As boundary conditions, we take $M(L) = 0$ (no seed producers at the end of the avalanche) and $n(0) = n_0$ (given initial seeds possibly due to a cosmic-ray source). A constant distributed cosmic-ray source S may be described by adding S/v_R to the right-hand side of the first equation in C1. This constant may be absorbed into $n' = n + S/(Rv_R)$ so the solution is now for n', with $n_0 = S/(Rv_R)$. From the first condition, we get $c_2 = -c_1(B/D)e^{DL}$, substituting which into the second condition, we find $c_1/D = n_0/\left(1 - \delta e^{DL}\right)$, where $\delta = BC/D^2$. We use the "smallness" of BC to assert $0 < \delta \ll 1$, which will eventually be confirmed by comparison with the results of Dwyer (2007).

Introducing

$$\gamma = \delta e^{DL} \tag{C2}$$

we have, for the RRE density at the end of avalanche ($x = L$):

$$n(L) = n_0 \frac{e^{RL}(1-\delta)}{1-\gamma} \approx \frac{n_0 e^{RL}}{1-\gamma}$$

By comparing to $n(L) = n_0 e^{RL}$ in the case of the absence of RFD mechanism, we conclude that, indeed, γ has the meaning of the feedback factor and is given by Equation C2, which is identical to Equation 6 when inserting the expression for D given above.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The model presented in Section 2, along with the code used to analyse and plot the results as seen in our figures, is available in Færder et al. (2025a), which also includes a text file with instructions on how to reproduce the results shown in the figures. Supplementary figures with additional results from our simulations are found in Færder et al. (2025b).

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